

# ON STALLINGS FIBRATION'S THEOREM FOR THE KNOT EXTERIORS

By

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# DECLARATION

This work was carried out at AIMS Rwanda in partial fulfilment of the requirements for a Master of Science Degree.

I hereby declare that except where due acknowledgement is made, this work has never been presented wholly or in part for the award of a degree at AIMS Rwanda or any other University.

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# DEDICATION

Veni, Vidi, Vici.

To Patricia DELHELLE, with gratitude for your love, support, and constant encouragement. This work is dedicated to you.

# Abstract

This thesis focuses on Stallings's fibration theorem in the context of knot exteriors. We introduce the fundamental tools of knot theory necessary for understanding fibred knots and the role of Seifert surfaces. After a brief review of algebraic topology and knot theory, we present the construction of the infinite cyclic covering, as well as Stallings' theorem and a proof of it, highlighting the deep connection between the algebraic structure of the knot's fundamental group and the existence of a fibration on the circle  $S^1$ .

**Keywords:** fibered knot, Seifert surface, knot complement, infinite cyclic covering, Stallings fibration theorem.

# Resume

**Titre en français:** Autour du théorème de fibration de Stallings pour les extérieurs de nœuds.

Ce mémoire porte sur le théorème de fibration de Stallings dans le cadre des extérieurs de nœuds. Nous y introduisons les outils fondamentaux de théorie des nœuds nécessaires à la compréhension des nœuds fibrés et du rôle des surfaces de Seifert. Après quelques rappels de topologie algébrique et de théorie des nœuds, nous présentons la construction du revêtement cyclique infini, ainsi que le théorème de Stallings et une démonstration de ce dernier, mettant en lumière le lien profond entre la structure algébrique du groupe fondamental du noeud et l'existence d'une fibration sur le cercle  $S^1$ .

**Mots-clés :** nœud fibré, surface de Seifert, complément de nœud, revêtement cyclique infini, théorème de fibration de Stallings.

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# 1. Introduction

## 1.1 Background and Motivation

The influence of Stallings' Fibration Theorem extends far beyond the scope of its original statement in [21]. This result has become a fundamental tool in the development of three-dimensional topology, establishing a deep connection between the algebraic properties of groups and the topology of fibrations of 3-manifolds. Its ideas have subsequently been developed and generalized in several different directions, particularly in low-dimensional topology.

One of the major developments stemming from this theory appears in Thurston's program on hyperbolic 3-manifolds. Wise's work on virtually special cube complexes, followed by Agol's proof of the Virtual Fibration Conjecture in [1], showed that a large class of 3-manifolds admits finite covers that fibre over the circle. These results rely on the interplay between the algebraic structure of the fundamental group and the topological properties of fibrations, a perspective for which Stallings' theorem constitutes one of the foundational starting points.

More recently, the ideas introduced by Stallings have continued to inspire new developments. In particular, Bridson, Kielak, and Kudlińska (2023) obtained a new homological proof of a version of Stallings' theorem in the setting of Poincaré duality groups and  $PD^3$ -pairs in [4]. Their work highlights the continuing relevance of Stallings' methods in the study of the relationship between algebraic properties of groups and the topological structures of manifolds.

Furthermore, fibering techniques remain ubiquitous in modern knot theory. They play an important role in the study of fibred knots, knot concordance, Seifert surfaces, and mapping torus structures. Thus, more than sixty years after its publication, Stallings' theorem remains a major reference at the interface of group theory, knot theory, and the topology of 3-manifolds.

## 1.2 Objectives of the thesis

The aim of this thesis is to present and prove Stallings' theorem for fibred knot complements, while illuminating throughout the chapters the dialogue between algebra and geometry that underlies it.

More specifically, this work aims to:

- briefly introduce the key concepts of classical knot theory necessary to carry out our study;
- present Seifert surfaces and their role in the topological structure of knot complements;
- construct the infinite cyclic covering associated with a knot complement and provide a geometric interpretation of it;
- present a proof of Stallings's fibration theorem in the context of knot exteriors;

## 1.3 Structure of the thesis

This thesis is organized into four chapters:

The second chapter presents the fundamental tools of algebraic topology and some key concepts of covering theory. In particular, we review some results from the geometry of 3-manifolds.

The third chapter serves as a brief introduction to knot theory; we develop the fundamental tools of this theory necessary for understanding this study. We describe the complement of a knot, notably its algebraic and geometric properties.

The fourth chapter is devoted to the construction of the infinite cyclic covering. Along the way, we present a proof of the structure theorem for the commutator subgroup of the knot group.

Finally, in the fifth chapter, we get to the heart of the matter. We present Stallings's fibration theorem along with a proof of it in the context of knot exteriors. Particular attention is paid to the Pretzel knots  $p(1, -1, 3)$ , which we use as examples to support the intuition behind Stallings's fibration theorem.

At the end of this journey, we hope to have convinced the reader that Stallings's theorem is much more than a technical result: it is a window onto the profound unity between topology and algebra.

## 2. PRELIMINARIES

This chapter is intended as a brief introduction to the algebraic and geometric tools that will serve as the foundation for the study of Stallings's fibration theorem for the exteriors of knots. We also establish the notation used throughout the thesis. We briefly review the essential concepts of algebraic topology and the topology of three-dimensional manifolds, as well as the important results concerning surfaces and fibered structures that will play a central role in this work. Most of the theorems stated here are accepted as established: we treat them as proven results and refer the reader to the appropriate references for more complete proofs.

### 2.1 Fundamental Group

**Definition 2.1.1** (Path, Loop). A *path*  $\gamma$  along a topological space  $X$  connecting  $x_0$  to  $x_1$  is a continuous mapping  $\gamma : I \longrightarrow X$  such that  $\gamma(0) = x_0$  and  $\gamma(1) = x_1$ .

In particular, if  $x_0 = x_1$ ,  $\gamma$  is called a *loop* in  $X$ .

In the following, we will denote by  $\Omega(X; x, y)$  the set of paths connecting  $x$  to  $y$  in  $X$ , and by  $\Omega(X; x_0)$ , the set of loops based at  $x_0$  in  $X$ .

**Definition 2.1.2** (Relative Homotopy [9, Definition 2.1, page 58]). Two paths  $\gamma_1$  and  $\gamma_2$  connecting the points  $x$  and  $y$  are said to be *homotopic relative to  $\{0, 1\}$*  if there exists a continuous map  $F : I \times I \longrightarrow X$  satisfying:

1.  $F|_{I \times \{0\}} = \gamma_0$  and  $F|_{I \times \{1\}} = \gamma_1$
2.  $F|_{\{0\} \times I} = x$  and  $F|_{\{1\} \times I} = y$

**Remark 2.1.3.** "Being homotopic relative to  $\{0, 1\}$ " defines an equivalence relation in space  $\Omega(X; -, -)$ .

We denote by  $\pi_1(X, x) = \Omega(X; x) / \simeq_{rel\{0, 1\}}$  the set of homotopy classes (relative to  $\{0, 1\}$ ) of loops with base point  $x$  in  $X$ .

**Definition 2.1.4.** Let  $\alpha$  and  $\beta$  be paths in  $X$  such that  $\alpha(1) = \beta(0)$ . We define the path

$$(\alpha \cdot \beta)(t) \begin{cases} \alpha(2t) & \text{if } 0 \leq t \leq \frac{1}{2}, \\ \beta(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1 \end{cases}$$

as the composed path of  $\alpha$  and  $\beta$ , denoted  $\alpha \cdot \beta$ .

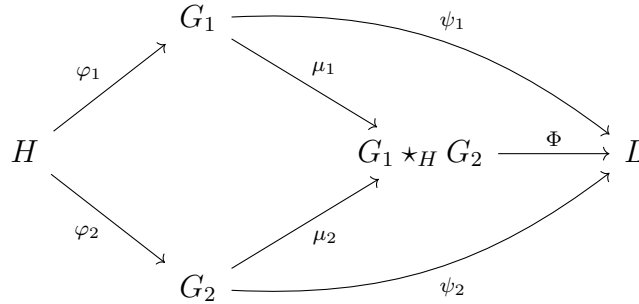
**Theorem 2.1.5** ([9, Theorem 2.2 page 75]). *The composition of paths induces a group structure on the set  $\pi_1(X, x)$  of homotopy classes of loops with base point  $x$  in  $X$ . The group  $\pi_1(X, x)$  is called the fundamental group of the topological space  $X$  at the point  $x$ .*

**Definition 2.1.6** (amalgamated sum [7, Definition 3.10]). Let  $\varphi_1 : H \rightarrow G_1$  and  $\varphi_2 : H \rightarrow G_2$  be group homomorphisms. Let  $N$  be the normal closure of the subgroup generated by the elements of the form  $\varphi_1(g) \varphi_2(g)^{-1}$  for all  $g \in H$ . The *amalgamated sum* of  $G_1$  and  $G_2$  over  $H$ , denoted  $G_1 \star_H G_2$ , is the quotient group  $(G_1 * G_2)/N$ , i.e.

$$G_1 \star_H G_2 := (G_1 * G_2) / \langle \overline{\varphi_1(g) \varphi_2(g)^{-1}} : g \in H \rangle$$

Let  $\varphi_1 : H \rightarrow G_1$  and  $\varphi_2 : H \rightarrow G_2$  be group homomorphisms. Let  $i_1 : G_1 \rightarrow G_1 * G_2$  and  $i_2 : G_2 \rightarrow G_1 * G_2$  denote the canonical injections into the free product. We denote by  $\mu_1 : G_1 \rightarrow G_1 \star_H G_2$  and  $\mu_2 : G_2 \rightarrow G_1 \star_H G_2$  the compositions of  $i_1$  and  $i_2$  with the canonical quotient map  $\pi : G_1 * G_2 \rightarrow G_1 \star_H G_2$ . By construction,  $\mu_1 \circ \varphi_1 = \mu_2 \circ \varphi_2$ .

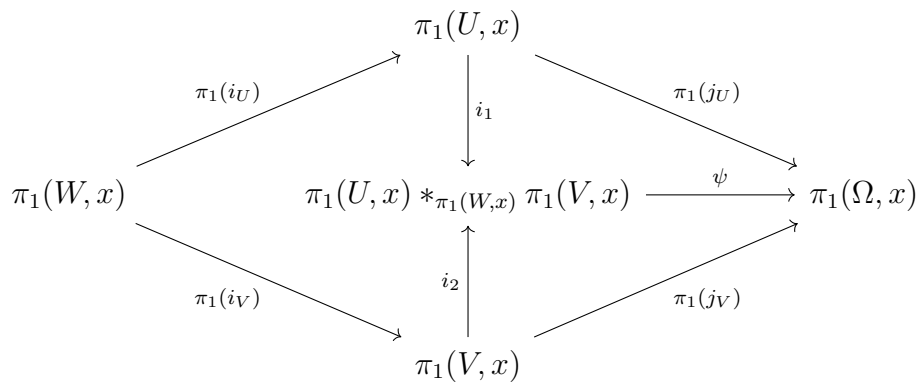
**Proposition 2.1.7** (Universal property of the amalgamated sum [7, Proposition 3.11]). Let  $\psi_1 : G_1 \rightarrow L$  and  $\psi_2 : G_2 \rightarrow L$  be group homomorphisms satisfying  $\psi_1 \circ \varphi_1 = \psi_2 \circ \varphi_2$ . Then there exists a unique group homomorphism  $\Phi : G_1 \star_H G_2 \rightarrow L$  such that  $\Phi \circ \bar{i}_1 = \psi_1$  and  $\Phi \circ \bar{i}_2 = \psi_2$ .



**Theorem 2.1.8** (Seifert–Van Kampen theorem[10, Theorem 2, page 30]). Let  $\Omega$  be a topological space equipped with a covering consisting of two open sets  $U$  and  $V$  such that:

- $U$  and  $V$  are arc-connected open sets,
- $W = U \cap V$  is an arc-connected open set.

The canonical injections  $i_U : U \cap V \rightarrow U$ ,  $i_V : U \cap V \rightarrow V$ ,  $j_U : U \rightarrow \Omega$ , and  $j_V : V \rightarrow \Omega$  induce the following commutative diagram



where  $x \in W$ , and  $\psi$  is a (group) isomorphism between  $\pi_1(\Omega, x)$  and the sum of  $\pi_1(U, x)$  and  $\pi_1(V, x)$  amalgamated by  $\pi_1(W, x)$ .

*Proof.* The existence of  $\psi$  follows from the universal property of the amalgamated sum. For a complete proof, we refer the reader to [7, Chapter 3, Theorem 3.12]  $\square$

**Example 2.1.9.** We compute the fundamental group of the unit two-sphere  $S^2$  using the Seifert–Van Kampen theorem. Let  $U = S^2 \setminus \{N\}$  and  $V = S^2 \setminus \{S\}$  be two opened subsets of the circle  $S^2$ , where  $N$  and  $S$  denote the north and south poles, respectively. Choose a base point  $x_0 \in U \cap V$ .

$U$  and  $V$  are homeomorphic to  $\mathbb{R}^2$ , hence contractible. It comes that  $\pi_1(U, x_0) = \pi_1(V, x_0) = \{1\}$ .

Moreover,

$$U \cap V = S^2 \setminus \{N, S\} \cong S^1 \times (-1, 1),$$

and  $(-1, 1)$  is contractile, so

$$\pi_1(U \cap V, x_0) \cong \pi_1(S^1) \times \pi_1((-1, 1)) \cong \pi_1(S^1) \cong \mathbb{Z}.$$

Since  $U$ ,  $V$ , and  $U \cap V$  are path-connected and  $S^2 = U \cup V$ , the Seifert–Van Kampen theorem yields

$$\pi_1(S^2, x_0) \cong \pi_1(U, x_0) * \pi_1(V, x_0) / N,$$

where  $N$  is the normal subgroup generated by the images of  $\pi_1(U \cap V, x_0)$  under the inclusion-induced homomorphisms.

Since  $\pi_1(U, x_0) = \pi_1(V, x_0) = \{1\}$ , we obtain

$$\pi_1(S^2, x_0) \cong \{1\} * \{1\} = \{1\}.$$

Finally,

$$\boxed{\pi_1(S^2) = \{1\}}.$$

**Proposition 2.1.10** ([9, Proposition 2.3, page 76]). Let  $\eta$  denote a path connecting a point  $x$  to another point  $y$  in  $X$ . The map  $\phi_\eta : [\gamma] \mapsto [\eta \gamma \bar{\eta}]$  is an isomorphism from the group  $\pi_1(X, y)$  to the group  $\pi_1(X, x)$  that depends only on the homotopy class of  $\eta$ .

We conclude this subsection by stating three results that will be of great use to us in the remainder of our study.

**Proposition 2.1.11.** Let  $f : G \longrightarrow H$  be an isomorphism and consider the induced homomorphism  $\bar{f} : G/[G, G] \longrightarrow H/[H, H]$ . In particular, this homomorphism defined as  $\bar{f}([g]) = [f(g)]$  is an isomorphism (of groups).

*Proof.* We show that  $\bar{f}$  is a well-defined homomorphism, injective, and surjective.

- Well-defined homomorphism: Recall that any commutator has the form  $[x, y] = xyx^{-1}y^{-1}$ , and for any homomorphism  $h$  we have

$$h([x, y]) = h(xyx^{-1}y^{-1}) = h(x)h(y)h(x)^{-1}h(y)^{-1} = [h(x), h(y)].$$

Since  $f$  is an isomorphism (in particular surjective), it follows that  $f([G, G]) = [H, H]$ . By the standard argument,  $\bar{f}$  is then well defined (independent of the choice of coset representative) and is a homomorphism, since  $f$  is one.

- Injectivity: Let  $[g] \in G/[G, G]$  such that  $\bar{f}([g]) = [f(g)] = [e_H]$ , i.e.  $f(g) \in [H, H]$ . Since  $f([G, G]) = [H, H]$  and  $f$  is injective, this forces  $g \in [G, G]$ , so  $[g] = [e_G]$ . Hence  $\ker \bar{f} = \{[e_G]\}$  and  $\bar{f}$  is injective.
- Surjectivity: Let  $[h] \in H/[H, H]$ . Since  $f$  is surjective, there exists  $g \in G$  with  $f(g) = h$ , so  $\bar{f}([g]) = [f(g)] = [h]$ . Hence  $\bar{f}$  is surjective.

Therefore  $\bar{f}$  is a bijective homomorphism, then it is an isomorphism.  $\square$

Let  $\gamma$  be a path along a topological space  $X$ ; we associate a singular 1-simplex,  $h(\gamma) : \Delta^1 \rightarrow X$ , given by  $h(\gamma)(te_1 + (1-t)e_2) = \gamma(t)$ .

**Theorem 2.1.12** (Hurewicz's Theorem [7, Theorem 7.1]). *For any arc-connected space  $X$ , the correspondence  $\gamma \mapsto h(\gamma)$  generates a group morphism,  $h_* : \pi_1(X) \rightarrow H_1(X; \mathbb{Z})$ , called the Hurewicz morphism. It induces an isomorphism between abelianization of  $\pi_1(X)$  and the group  $H_1(X; \mathbb{Z})$ , i.e.  $H_1(X; \mathbb{Z}) \cong \pi_1(X)/[\pi_1(X), \pi_1(X)] = \text{Ab}(\pi_1(X))$ .*

*Proof.* For technical details regarding this proof, see [7, Chapter 7, Theorem 7.1, p.198].  $\square$

The scope of this thesis focuses on the development of the third dimension; therefore, we will take the classification of surfaces as given. Readers wishing to expand their knowledge on the subject are strongly encouraged to explore Andre Gramain's book, [10] Chapter 5, pages 74–92. We recall the homology of a surface.

For an integer  $g \geq 1$ , let  $T_g := \#^g(S^1 \times S^1)$  denote the torus with  $g$  holes. We denote  $V_g$  as the torus with  $g$  holes minus a small disk, i.e.,  $V_g := \#^g(S^1 \times S^1) \setminus \text{int}(D^2)$  with  $D^2 = \{x \in \mathbb{R}^2 \mid |x| \leq 1\}$ .

**Theorem 2.1.13.** *Every compact, connected, orientable surface  $M$  of genus  $g \geq 1$  is homeomorphically (abstractly) a connected sum of  $g$  tori, i.e.,  $M \cong T_g := \#^g(S^1 \times S^1)$*

*Proof.* See [10, Chapter 5, Theorem 4, p.86].  $\square$

**Proposition 2.1.14.** The fundamental group of the space  $V_g$  is a free group with  $2g$  generators  $a_1, b_1, \dots, a_g, b_g$ . The boundary  $\gamma$  of  $T_g \setminus \text{Int}(D^2)$  has class  $\bar{\gamma} = [a_1, b_1][a_2, b_2] \cdots [a_g, b_g]$ . Furthermore, the fundamental group of  $T_g$  is isomorphic to the quotient of the former by the normal subgroup generated by  $\bar{\gamma}$ . In other words,



$$\pi_1(V_g) = \mathbb{F}_{2g} \text{ and } \pi_1(T_g) \cong \pi_1(V_g) / \langle \bar{\gamma} \rangle = \mathbb{F}_{2g} / \langle \bar{\gamma} \rangle.$$

where  $\mathbb{F}_{2g}$  denotes the free group with  $2g$  generators.

*Proof.* See [10, Chapter 2, Section 6, p.86]. □

## 2.2 Covering spaces

Intuitively, a covering of a topological space  $X$  consists of opening some non-trivial loops of that space. We will assume that  $X$  is an arc-connected and locally arc-connected<sup>1</sup> topological space.

**Definition 2.2.1** (Covering space [7, Definition 4.1]). A *covering space* of a space  $X$  is a space  $\tilde{X}$  arc-connected and locally arc-connected, together with a continuous map  $p : \tilde{X} \rightarrow X$  satisfying the following condition: every point  $x \in X$  has a connected neighborhood,  $U$ , such that  $p^{-1}(U)$  is a disjoint union of open sets  $U_i$ , and the restriction  $p|_{U_i} : U_i \rightarrow U$  is a homeomorphism.

**Example 2.2.2.** Let  $n$  be a non-zero natural integer.

- In the diagram on the right<sup>2</sup> of Figure 2.1, making one complete revolution on the top circle is equivalent to making  $n$  revolutions on the bottom circle: We define the  $n$ -order covering on the circle given by  $p_n : S^1 \rightarrow S^1, z \mapsto z^n$ .
- In the infinite version (the diagram on the left) of Figure 2.1: making one turn in  $\tilde{X}$  at the top is equivalent to making one turn of the circle at the bottom. Looking more closely at the figure, we notice that  $\tilde{X} = \mathbb{R}$ : We define a covering on the circle given by  $p : \mathbb{R} \rightarrow S^1, t \mapsto e^{2i\pi t}$ . In particular, it is often called the universal covering on  $S^1$  because its total space  $\mathbb{R}$  is simply connected<sup>3</sup>.

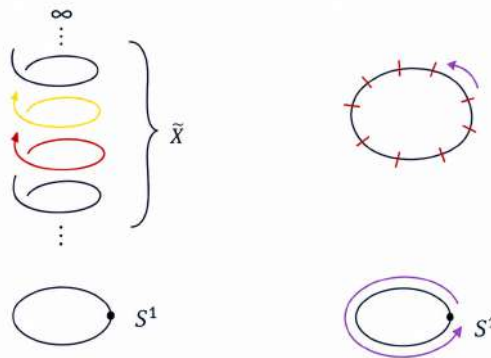


Figure 2.1: The different coverings on the circle  $S^1$ .

<sup>1</sup>a space with a basis of arc-connected neighborhoods

<sup>2</sup>the top circle represents a circle divided into  $n$  parts

<sup>3</sup>connected by arcs and  $\pi_1(\tilde{X}) = \{1\}$

**Definition 2.2.3** ([9, Definition 4.3 page 135]). Let  $p : \widetilde{X} \rightarrow X$  be a covering,  $x \in X$ , and  $\tilde{x} \in p^{-1}(x)$ . The covering  $p$  is *Galoisian* (or *regular*) if the subgroup  $p_*\pi_1(\widetilde{X}, \tilde{x})$  is normal in  $\pi_1(X, x)$ .

**Example 2.2.4.** Every universal covering is a regular covering. Indeed,  $\widetilde{X}$  being a universal covering implies that  $\pi_1(\widetilde{X}) = \{1\}$ . Thus  $p_*\pi_1(\widetilde{X}) = \{1\}$ , which is a distinct element in the group  $\pi_1(X)$ .

**Theorem 2.2.5** (Characterization of regular coverings [7, Theorem 4.27]). *Let  $p : \widetilde{X} \rightarrow X$  be a covering. We have the following equivalence:*

1. *The covering  $p$  is regular.*
2. *The automorphism group,  $\text{Aut}(p)$ , acts transitively on the fiber  $p^{-1}(x)$ , for all  $x \in X$ .*

The following result establishes a deep connection between the subgroups of the fundamental group and the coverings of a topological space.

**Theorem 2.2.6** (Classification of coverings [7, Cor 4.36]). *Let  $X$  be an arc-connected topological space. The isomorphism classes of coverings of  $X$  are in bijection with the conjugacy classes of subgroups of  $\pi_1(X)$ , with regular coverings corresponding to normal subgroups. In other words, the following correspondences are bijective:*

$$\left\{ \text{The coverings of } X \right\} \longleftrightarrow \left\{ \text{The subgroups of } \pi_1(X) \right\}$$

And

$$\left\{ \text{The regular coverings of } X \right\} \longleftrightarrow \left\{ \text{The normal subgroups of } \pi_1(X) \right\}$$

**Remark 2.2.7.** In particular, if  $X$  is simply connected, then  $X$  is the only covering of itself.

In this work, where the main space we are dealing with is the exterior of the knot, we will focus on the characteristic subgroups of the fundamental group of the knot: it follows from Theorem 2.2.6 that there is a geometric construction directly linked to the existence of a characteristic subgroup, which we will take care to define precisely.

## 2.3 Locally trivial fibers

**Definition 2.3.1** ([7, Definition 7.6]). A continuous map  $p : E \rightarrow B$  is a *fibration* if, for the topological space  $X$  and for every commutative diagram of continuous maps,

$$\begin{array}{ccc}
X \times \{0\} & \xrightarrow{f_0} & E \\
j_0 \downarrow & \nearrow \tilde{f} & \downarrow p \\
X \times I & \xrightarrow{f} & B
\end{array}$$

there exists a continuous map  $\tilde{f} : X \times I \rightarrow E$  such that  $p \circ \tilde{f} = f$  and  $\tilde{f} \circ j_0 = f_0$ .

**Definition 2.3.2** ([7, Definition 7.11]). A *fiber bundle* with base  $B$  and fiber  $F$  is a continuous map  $p : E \rightarrow B$  satisfying the following property: there exists an open cover  $(U_i)_{i \in I}$  of  $B$  and homeomorphisms  $\Phi_i : p^{-1}(U_i) \rightarrow U_i \times F$ , such that  $p_{U_i} \circ \Phi_i = p$ .

$$\begin{array}{ccc}
p^{-1}(U_i) & \xrightarrow{\Phi_i} & U_i \times F \\
p \searrow & & \swarrow p_{U_i} \\
& U_i &
\end{array}$$

**Example 2.3.3.** ■ The torus  $T_2 = S^1 \times S^1$  equipped with the first canonical (continuous!) projection  $\text{proj}_1 : T_2 \rightarrow S^1$ ,  $\text{proj}_1(x, y) = x$  is a fiber bundle over the circle  $S^1$ . Indeed, the pair  $\{U_1, U_2\}$  with  $U_1 = S^1 \setminus \{(0, 1)\}$  and  $U_2 = S^1 \setminus \{(0, -1)\}$  represents an open cover of  $S^1$ .

For every  $x \in S^1$ , we have  $\text{proj}_1^{-1}(x) = \{x\} \times S^1 \cong S^1$ . In other words, the fiber over each point of the circle is homeomorphic to  $S^1$ .

Furthermore, for any  $i \in \{1, 2\}$ , it is easy to verify that  $\text{proj}_1^{-1}(U_i) \cong U_i \times S^1$ : it suffices to consider the homeomorphisms  $\varphi_i : \text{proj}_1^{-1}(U_i) \rightarrow U_i \times S^1$ , where  $\varphi_i(x, y) = (x, y)$  and their inverses are  $\varphi_i^{-1}(x, y) = (x, y)$ . And we obtain the following commutative diagram:

$$\begin{array}{ccc}
\text{proj}_1^{-1}(U_i) & \xrightarrow{\cong} & U_i \times S^1 \\
\text{proj}_1|_{U_i} \searrow & & \swarrow p_{U_i} \\
& U_i &
\end{array}$$

It follows that  $\text{proj}_1 : T_2 \rightarrow S^1$  is a fiber bundle over  $S^1$ . In particular, this fiber bundle is even globally trivial.

- The Klein bottle as a non-trivial fiber bundle over the circle

Consider the Klein bottle defined by

$$K = \frac{[0, 1] \times S^1}{(0, z) \sim (1, \bar{z})}.$$

and the projection  $p : K \rightarrow S^1$ ,  $p([t, z]) = e^{2\pi it}$ , which is a fiber bundle with fiber  $S^1$ .

Indeed, for every open set  $U \subset S^1$  we have  $p^{-1}(U) \cong U \times S^1$ . Thus,  $p$  is a locally trivial bundle.

Moreover, for every point  $x \in S^1$ , the fiber is  $p^{-1}(x) \cong S^1$ .

Let us show that this bundle is not globally trivial. Suppose that  $K \cong S^1 \times S^1 = T_2$ . It follows that their fundamental groups are isomorphic.

However,

$$\pi_1(K) = \langle a, b \mid aba^{-1} = b^{-1} \rangle,$$

whereas

$$\pi_1(T_2) = \langle a, b \mid ab = ba \rangle \cong \mathbb{Z}^2.$$

We observe that the two groups are not isomorphic. We deduce that

$$K \not\cong T_2.$$

Consequently, the bundle  $p : K \rightarrow S^1$  is a locally trivial fiber bundle with fiber  $S^1$ . This bundle is not (globally) trivial.

**Theorem 2.3.4** ([7, Theorem 7.13]). *Every fiber bundle is a fibration.*

*Proof.* The proof is quite technical; therefore, we refer the reader to a detailed proof in [7], Theorem 7.13, p.248.

□

**Remark 2.3.5.** The concept of a fibration generalizes that of a fiber bundle. As we will see later, certain 3-manifolds have a fibration structure over the circle  $S^1$ . This idea plays a role in Stallings' theorem.

We conclude this subsection with the definition of a geometric object that will be of great use to us in the remainder of our study for capturing the geometric representation of certain objects.

**Definition 2.3.6** (Handlebody, [5, Definition 3.20]). A *handlebody*  $M$  of genus  $g$  is a space obtained from a 3-ball  $\mathbb{B}^3$  by attaching  $g$  handles  $D^2 \times I$  such that the boundary  $\partial M$  is an orientable closed surface of genus  $g$ :  $M = \mathbb{B}^3 \cup H_1 \cup \cdots \cup H_g$ ,  $H_i \cap H_j = \emptyset$  ( $i \neq j$ ),

$H_i \cap \mathbb{B}^3 = D_{i1} \cup D_{i2}$ ,  $D_{i1} \cap D_{i2} = \emptyset$ ,  $D_{ij} \cong D^2$  and  $\left( \partial \mathbb{B}^3 \setminus \bigcup_{i,j} D_{ij} \right) \cup \bigcup_i (\partial H_i \setminus (D_{i1} \cup D_{i2}))$

is a closed orientable surface of genus  $g$ .

**Example 2.3.7.**

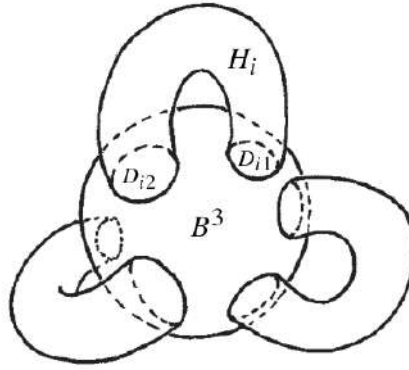


Figure 2.2: A 3-handlebody

We denote by  $U(D^2)$ , the closed regular neighbourhood of  $D^2$ , that is  $U(D^2) \cong D^2 \times I$ .

**Proposition 2.3.8** ([5, Proposition 3.21]). Let  $W$  be an orientable 3-manifold. If  $W$  contains a system  $D_1, \dots, D_g$  of mutually disjoint disks such that  $\partial W \cap D_i = \partial D_i$  with  $U(D_i) \cap U(D_j) = \emptyset$  for  $i \neq j$ , and  $W \setminus \bigcup_i U(D_i)$  is a closed 3-ball, then  $W$  is a handlebody of genus  $g$ .

The results in this chapter form the foundation upon which the rest of the course will be built. The next chapter dives right into knot theory, exploring the main properties of their complements.

# 3. The basics of knot theory and the complement of a knot

It would be premature to address the statement of Stallings' theorem without first introducing the concept of knots. This chapter therefore provides an overview of knot theory, presenting its fundamental tools as well as some significant results. We refer readers wishing to deepen their knowledge of the subject to the literature. The homology we deal with in this chapter is singular homology with integer coefficients.

## 3.1 Some basic definitions

The following results are taken from Burde and Zieschang book [5], to which we refer the reader for further details.

**Definition 3.1.1.** A knot is a topological embedding  $\mathfrak{k} : S^1 \hookrightarrow \mathbb{R}^3$  (or  $S^3$ ).

**Remark 3.1.2.** Intuitively, a knot can simply be visualized as the image  $\mathfrak{k}(S^1)$ : Indeed, the circle  $S^1$  is the abstract object we are working with, but it is the way it is arranged in three-dimensional space (or  $S^3$ ) that represents a knot.

**Definition 3.1.3** (Regular projection). A projection  $p$  of a knot  $\mathfrak{k}$  is called *regular* if there are only finitely many multiple points  $\{P_i \mid 1 \leq i \leq n\}$ , and all multiple<sup>1</sup> points are *double points*, that is,  $p^{-1}(P_i)$  contains two points of  $\mathfrak{k}$ .

**Theorem 3.1.4.** A knot  $\mathfrak{k}$  always has a regular projection.

**Definition 3.1.5.** A *diagram* is a regular projection of a knot where height information is indicated at each double singular point. By convention, the dashed line corresponds to the lower segment.

**Example 3.1.6.** Figure 3.1 represents a diagram of the pretzel knot  $p(4, 4, 4)$ .

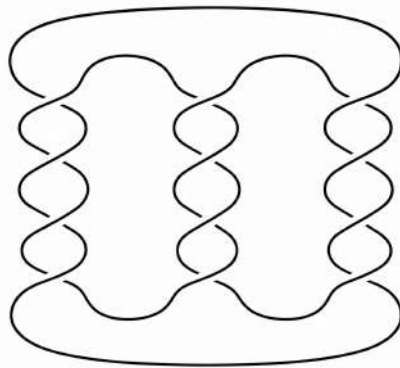


Figure 3.1: Diagram of the pretzel knot  $p(4, 4, 4)$

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<sup>1</sup>A point  $x$  in the plane is called a multiple point of the projection  $p$  if the fiber  $p^{-1}(x) \cap \mathfrak{k}$  contains more than one point.

Since a regular projection has a finite number of double singular points, it follows that a diagram has a finite number of crossings.

**Theorem 3.1.7.** *Every knot has a diagram.*

*Proof.* This follows from the definition 3.1.4 and the theorem 3.1.5. □

## 3.2 Reidemeister moves

We now know that a knot is an object that exists in  $S^3$  (or in  $\mathbb{R}^3$ ), which is manipulated on a diagram via a regular planar projection. However, for a given knot, there may be several (or even an infinite number) of diagrams associated with it. From here on, we may ask ourselves: When do two diagrams represent the same knot?

Figures 3.2, 3.3, and 3.4 illustrate the three elementary Reidemeister moves, which allow us to modify (locally) a diagram.

- The Reidemeister move of type I consists of creating or removing a loop.



Figure 3.2: Reidemeister moves of type I

- The Reidemeister moves type II involves inserting or removing two crossings.

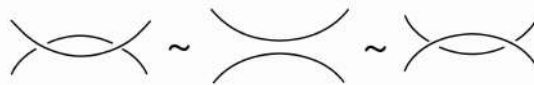


Figure 3.3: Reidemeister moves of type II

- The Reidemeister moves type III consists of passing a strand of the diagram over an existing crossing.



Figure 3.4: Reidemeister moves of type III

**Theorem 3.2.1** (Reidemeister, [5]). *Two knot diagrams are called equivalent if they are connected by a finite sequence of Reidemeister moves  $\Omega_i$ ,  $i = 1, 2, 3$  or their inverses  $\Omega_i^{-1}$ .*

*Proof.* The idea behind the proof of this theorem rests on the fact that two regular projections of the same knot that deform continuously (from one to the other and vice versa) change the diagram only when passing through singular projections. For more details, see the proof of Lemma 1.14 on p. 9 of [5].  $\square$

### 3.3 Knot group

For the moment, let  $V(\mathfrak{k})$  be a closed tubular neighbourhood of the knot  $\mathfrak{k}$ . A precise definition of a tubular neighbourhood will be given later in section 3.5.1.

**Definition 3.3.1.** The *knot group* of the knot  $\mathfrak{k}$  is the fundamental group of the complement of the knot  $\pi_1(C) = \pi_1(S^3 \setminus V(\mathfrak{k}))$ .

An effective technique developed by the mathematician Wirtinger<sup>2</sup> allows us to determine a presentation of the fundamental group of the complement of a knot from its (oriented) diagram.

**3.3.2 Wirtinger's presentation.** To begin, let us consider a point  $x_0$  in the complement of the knot. Intuitively, we know that evaluating  $\pi_1(C, x_0)$  amounts to identifying all loops (based at  $x_0$ ) that are non-contractile in the space  $C$ . Now, consider a loop  $s_i$  based at  $x_0$  that bounds a disk  $D_i$ , and that transversely intersects the knot  $k$  at exactly one point for each of the strands  $b_i$  of our knot, so that the loop  $s_i$  links once around the disk  $D_i$ . The idea is that to each arm of the oriented diagram<sup>3</sup>  $b_i$ , we associate a generator of the fundamental group that is none other than  $s_i$ , which is homotopic to the meridian. We still need to understand what happens at the crossings, i.e., whether there are any relations between these loops  $s_j$ : From Figure 3.5, we can observe that each crossing naturally induces a relation among the loops corresponding to the three strands meeting at that crossing.

<sup>2</sup>Whose real name was Wilhelm Wirtinger (1865–1945), an Australian mathematician

<sup>3</sup>The fundamental group does not depend on the chosen orientation, but it should be noted that the orientation plays an important role in specifying any generators



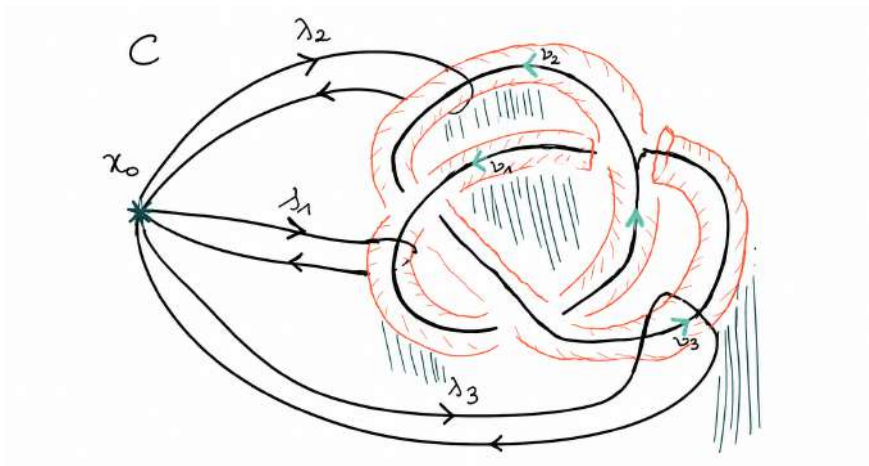


Figure 3.5: Generators of the knot group

Suppose that an arc  $s_k$  passes above two consecutive arcs  $s_i$  and  $s_{i+1}$ . Depending on the orientation of the crossing, two cases may occur:

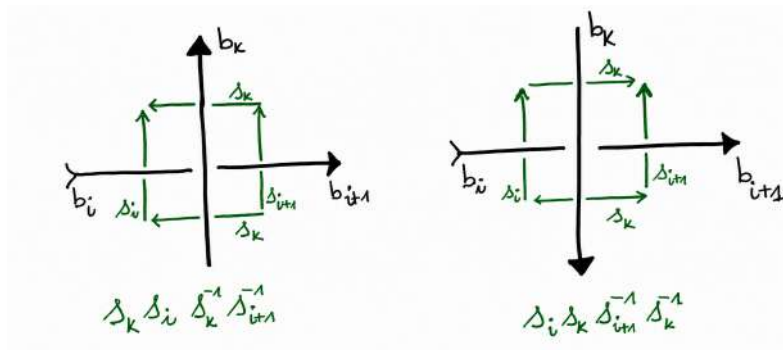


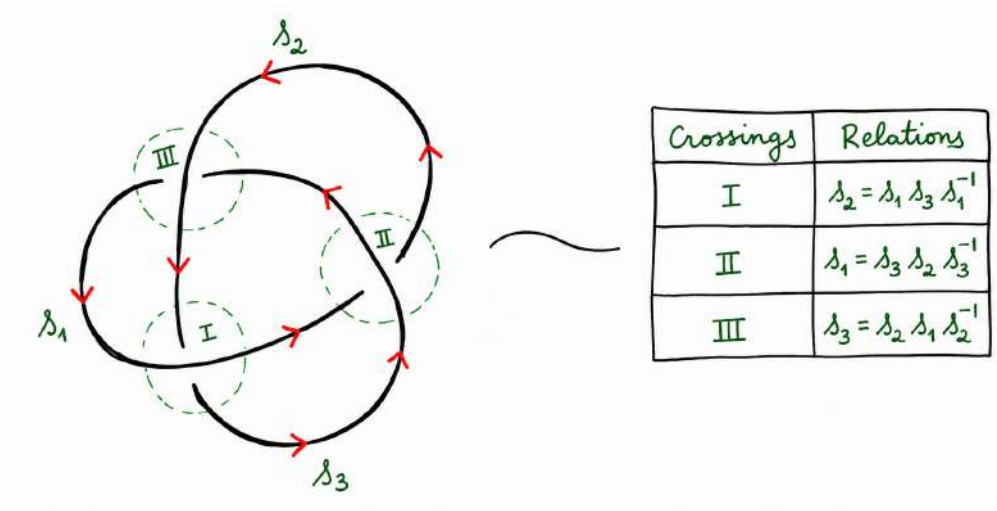
Figure 3.6: Wirtinger's relations

These relations reflect the effect of passing under an arc on the corresponding loops.

**Theorem 3.3.3** (Wirtinger presentation [5, Theorem 3.4]). *Let  $s_i$ ,  $i = 1, 2, \dots, n$ , be the overcrossing arcs of a regular projection of a knot  $\mathfrak{k}$ . Then the knot group admits the following so-called Wirtinger presentation:*

$$\mathfrak{G} = \pi_1(\overline{S^3 \setminus V(\mathfrak{k})}) = \langle s_1, \dots, s_n \mid \text{relations at each intersection of the diagram} \rangle.$$

**Example 3.3.4.** (Presentation of the trefoil knot group =  $3_1$ )

Figure 3.7: The knot group of  $3_1$ 

From Theorem 3.3.3, we obtain:

$$\pi_1(\overline{S^3 \setminus V(3_1)}) = \langle s_1, s_2, s_3 \mid s_1 s_3 s_1^{-1} s_2^{-1}, s_3 s_2 s_3^{-1} s_1^{-1}, s_2 s_1 s_2^{-1} s_3^{-1} \rangle$$

It is possible to reduce the resulting presentation to two generators. Indeed, in the Wirtinger presentation above, we have:  $s_2 = s_1 s_3 s_1^{-1}$  and  $s_1 = s_3 s_2 s_3^{-1}$ .

Substituting  $s_2$  into the second relation, we obtain:  $s_1 = s_3 (s_1 s_3 s_1^{-1}) s_3^{-1} = s_3 s_1 s_3 s_1^{-1} s_3^{-1}$ . This last relation implies  $s_1 s_3 s_1 = s_3 s_1 s_3$  (\*)

The new representation becomes,

$$\pi_1(\overline{S^3 \setminus V(3_1)}) = \langle s_1, s_3 \mid s_3 s_1 s_3 = s_1 s_3 s_1 \rangle.$$

We can go further by setting  $x = s_1 s_3 s_1$  and  $y = s_1 s_3$ , the relation  $s_3 s_1 s_3 = s_1 s_3 s_1$  is equivalent to  $x^2 = y^3$ . Indeed, by using the previous relation (\*) we have  $y^3 = s_1 s_3 s_1 s_3 s_1 s_3 = s_1 s_3 s_1 s_1 s_3 s_1 = x^2$ .

Finally, we obtain:

$$\pi_1(\overline{S^3 \setminus V(3_1)}) = \langle x, y \mid x^2 = y^3 \rangle.$$

**Remark 3.3.5.** It is always possible to rewrite one of the  $n$  relations as a consequence of the other  $n - 1$ , so that it is in fact redundant. More generally, we have the following proposition

**Proposition 3.3.6.** Let  $\mathfrak{k}$  be a knot and  $\langle s_1, \dots, s_n \mid r_1, \dots, r_n \rangle$  a Wirtinger presentation of  $\mathcal{G}$ . Then each defining relation  $r_j$  is a consequence of the other defining relations  $r_i, i \neq j$ .

*Proof.* For a detailed description, see [5, Corollary 3.6]

□

### 3.4 Seifert surfaces

The results in this section are largely based on the book by Rolfsen [17], to which we refer the reader for the proofs.

**Definition 3.4.1.** Every simple closed curve  $k \subset \mathbb{R}^3$  bounds a orientable surface<sup>4</sup> embedded in  $\mathbb{R}^3$ , called a *Seifert surface*<sup>5</sup>.

**Consequence 3.4.2.** Every knot is the boundary of a Seifert surface.

*Proof.* Rigorously proving the previous proposition within the scope of this thesis may seem very complicated but an alternative relies on an algorithm by Seifert, which allows one to find a Seifert surface from a given knot diagram □

**3.4.3 Construction of Seifert surfaces. Seifert's algorithm:** Let  $D$  be an oriented diagram of a knot  $N$ . The construction proceeds as follows:

1. Choose a starting point on an arc of the diagram.
2. Traverse the diagram following the orientation. At each crossing, move to the adjacent arc so as to preserve consistency with the knot's orientation.

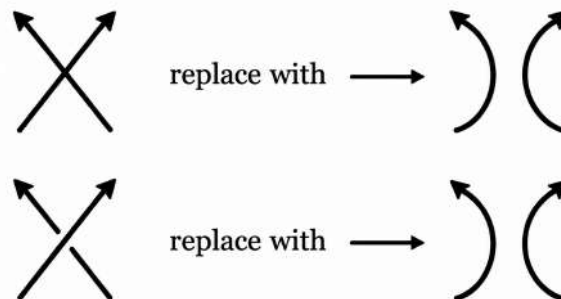


Figure 3.8: Orientation

3. This yields a finite family of disjoint closed curves, called Seifert circles.
4. Each Seifert's circle bounds a disk, which we can assume is embedded in distinct planes so that the disks are disjoint.

<sup>4</sup>Any differentiable manifold of dimension two.

<sup>5</sup>Herbert Seifert (1907–1996) was a German mathematician.

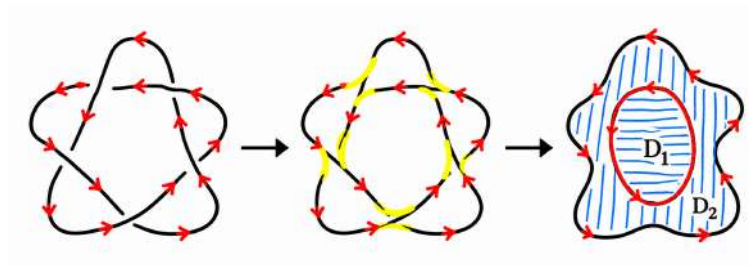


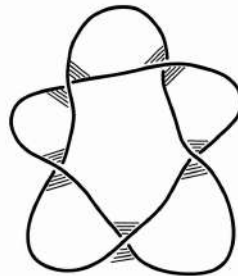
Figure 3.9: step 1+ step 2 + step 3

5. For each initial intersection in the diagram  $D$ , connect the corresponding disks with a twisted band, chosen according to the type of intersection.



Figure 3.10: twisted bands

6. This yields a Seifert surface of the knot. And that's all!

Figure 3.11: Seifert surface of the knot  $5_1$ 

### 3.4.4 Genus of a knot.

**Definition 3.4.5** (Genus). The *genus* of the knot  $\mathfrak{k}$  is the minimal genus  $g$  of a Seifert surface spanning the knot  $\mathfrak{k}$ . In other words, the minimal genus of a knot  $\mathfrak{k}$ , denoted  $g(\mathfrak{k})$ , is given by:

$$g(\mathfrak{k}) = \min \left\{ g(S) \mid S \text{ is a Seifert surface of the knot } \mathfrak{k} \right\}$$

**Example 3.4.6.** ■ The trivial knot bounds a disk that is its Seifert surface (i.e.,  $\partial D^2 = S^1$  with  $D = \{x \in \mathbb{R}^2 \mid |x|_{\mathbb{R}^2} < 1\}$  and  $S = D$ ). Since the disk is not perforated, it follows that  $g(\text{trivial knot}) = 0$ .

- La figure 3.11 shows that the clover knot has a Seifert surface of genus 2. However, it cannot bound a surface of genus zero, i.e., a disk, because that would imply that it is trivial, which is false.

**Remark 3.4.7.** It should be noted, however, that calculating the genus of a knot can be tricky. Indeed, the genus of the surface obtained by Seifert's algorithm depends on the diagram considered, and furthermore, this algorithm does not necessarily produce a surface of minimal genus.

**3.4.8 linking Number.** Let  $j$  and  $k$  be two disjoint oriented knots. There are several ways to define the linking number. For the purposes of this study, we prefer a homological version because it seems much more elegant and intuitive.

**Definition 3.4.9.** Using the previous notation, if we denote  $S_j$  as the Seifert surface of the knot  $j$ , the linking number is defined as:

$$\text{lk}(j, k) = \text{lk}([j], [k]) := [S_j] * [k]$$

where  $[S_j] * [k]$  denotes the (algebraic) number of intersections between  $k$  and the surface  $S_j$ .

Intuitively, each time the knot  $k$  crosses the surface  $S_j$ , we count  $\pm 1$  depending on the different orientations of  $S_j$  and  $k$ : the sum of these signs denotes the linking number  $\text{lk}(j, k)$ .

Hereinafter,  $k$  denotes an oriented knot.

**3.4.10 Positive push-off and Seifert form.** Let  $S_k$  be a Seifert surface of the knot  $k$ . Since it is orientable (by definition), there exists a non-zero field of normal vectors  $X$  along the surface  $S_k$ . Now, consider  $\eta$  a simple oriented curve on the surface  $S_k$ .

**Definition 3.4.11.** The positive push-off of the curve  $\eta$ , denoted  $\eta^+$ , is a curve parallel<sup>6</sup> to  $\eta$  along the complement of  $S_k$ . In other words,

$$\begin{aligned} \eta^+ : I &\rightarrow S^3 \setminus S_k \\ t &\mapsto \eta^+(t) = \eta(t) + \epsilon X(\eta(t)) \end{aligned}$$

for some sufficiently small  $\epsilon > 0$

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<sup>6</sup> $\eta^+$  represents a slightly shifted copy of the curve  $\eta$  in the complement of the surface.

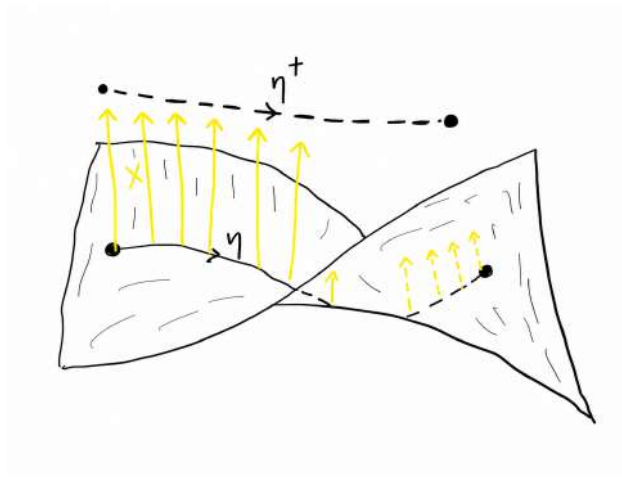


Figure 3.12: positive push-off

**Example 3.4.12.** The curve  $\alpha^+$  denotes the positive push-off of  $\alpha$ , obtained by moving  $\alpha$  slightly along the positive (green color) normal direction to the Seifert surface.

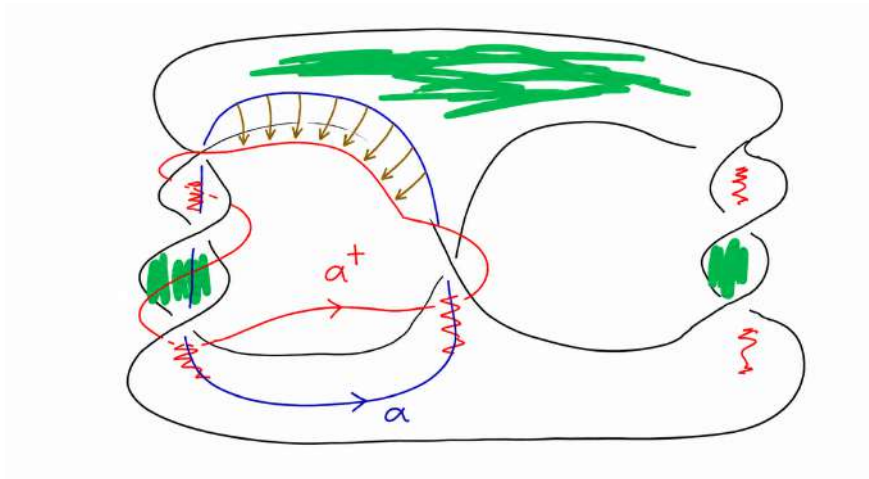
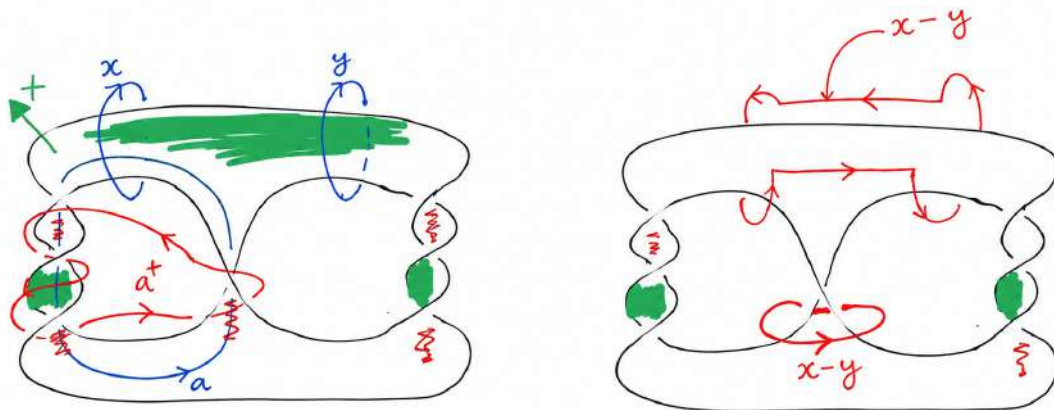


Figure 3.13: Illustration of the positive push-off

Moreover, The figure 3.14 shows that the positive push-off  $a^+$  is homologous on the Seifert surface to a curve obtained by traversing the generator  $x$  twice and the generator  $y$  once in the opposite direction. Therefore, with respect to the canonical basis  $(x, y)$  of  $H_1(S)$ , we have

$$a^+ = 2x - y.$$


 Figure 3.14: Expression in the canonical base of  $\pi_1$ 

**Definition 3.4.13.** The Seifert form is the mapping defined by,

$$\begin{aligned} \Xi_k : H_1(S_k) \times H_1(S_k) &\rightarrow \mathbb{Z} \\ ([\alpha], [\beta]) &\mapsto \Xi_k([\alpha], [\beta]) = \text{lk}(\alpha, \beta^+) \end{aligned}$$

**Proposition 3.4.14.** The map  $\Xi_k$  is  $\mathbb{Z}$ -bilinear.

*Proof.* La  $\mathbb{Z}$ -bilinéarité de la forme de Seifert  $\Xi_k$  découle du fait que le nombre d'enlacement est lui-même  $\mathbb{Z}$ -bilinéaire et que le push-off positif satisfait  $\eta^+(c+d) = c^+ + d^+$  et  $\eta^+(\mathbb{Z}a) = \mathbb{Z}a^+$ .  $\square$

**3.4.15 Seifert Matrix.** Suppose  $H_1(S_k)$  is an abelian group with basis  $\{a_1, \dots, a_{2g}\}$  as we will see in 3.4.19, then the associated *Seifert matrix*  $V_k$  is the  $2g \times 2g$  square matrix with coefficients  $v_{ij} = \Xi_k([\alpha_i], [\alpha_j]) = \text{lk}(\alpha_i, \alpha_j^+)$  where  $\Xi_k$  is a Seifert form, and  $g$  is the genus of the Seifert surface  $S_k$ .

**3.4.16 From the perspective of surface classification.** We saw earlier that a Seifert surface is a compact, connected, orientable surface bounded by a knot; according to the theorem 2.1.13, it is homeomorphic to a torus with a disk removed. More precisely, we have the following proposition:

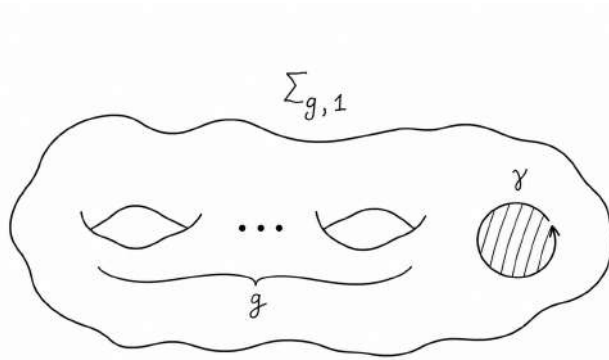
**Proposition 3.4.17.** A Seifert surface  $S$  of a knot  $k$  of genus  $g \geq 1$  is homeomorphic to a  $g$ -torus with a disk removed, i.e.,

$$S \cong \#^g(S^1 \times S^1) \setminus \text{Int}(D^2)$$

**Example 3.4.18.**


 Figure 3.15: abstract representation of the Seifert surface of  $5_1$ .

In general, we have the (abstract) representation:


 Figure 3.16: abstract representation of the Seifert surface of a knot of  $g$  .

**3.4.19 Fundamental Group and homology of a Seifert Surface.** In this subsection, we review some results on the calculation of fundamental groups of the fundamental group of a Seifert surface.

**Proposition 3.4.20.** The fundamental group of the torus with a hole,  $\#(S^1 \times S^1) \setminus \text{Int}(D^2)$ , is a free group with two generators

More generally, for  $g \geq 1$ , let  $T_g = \#^g(S^1 \times S^1)$  denote the torus with  $g$  holes. We denote  $V_g$  as the torus with  $g$  holes deprived of a small disk, i.e.,  $V_g = \#^g(S^1 \times S^1) \setminus \text{Int}(D^2)$ . We have the following theorem:

**Theorem 3.4.21.** A Seifert surface  $S$  of genus  $g$  is homeomorphic to  $V_g$ , so they have the same homotopy groups, i.e.  $\pi_1(S) = \pi_1(V_g) = \mathbb{F}_{2g}$ .

*Proof.* This follows from propositions 3.4.17 and 2.1.14. □



## 3.5 Around the complement of the knot

The following results are taken from book [18] to which we refer the reader for further details.

### 3.5.1 Tubular neighborhood of a knot.

**Theorem 3.5.2.** *A knot  $\mathfrak{k}$  has a tubular neighborhood, which is a neighborhood of  $\mathfrak{k}$ , denoted  $V(\mathfrak{k})$ , homeomorphic to  $S^1 \times D^2$  whose boundary  $\partial V(\mathfrak{k})$  is homeomorphic to the space  $S^1 \times S^1$ .*

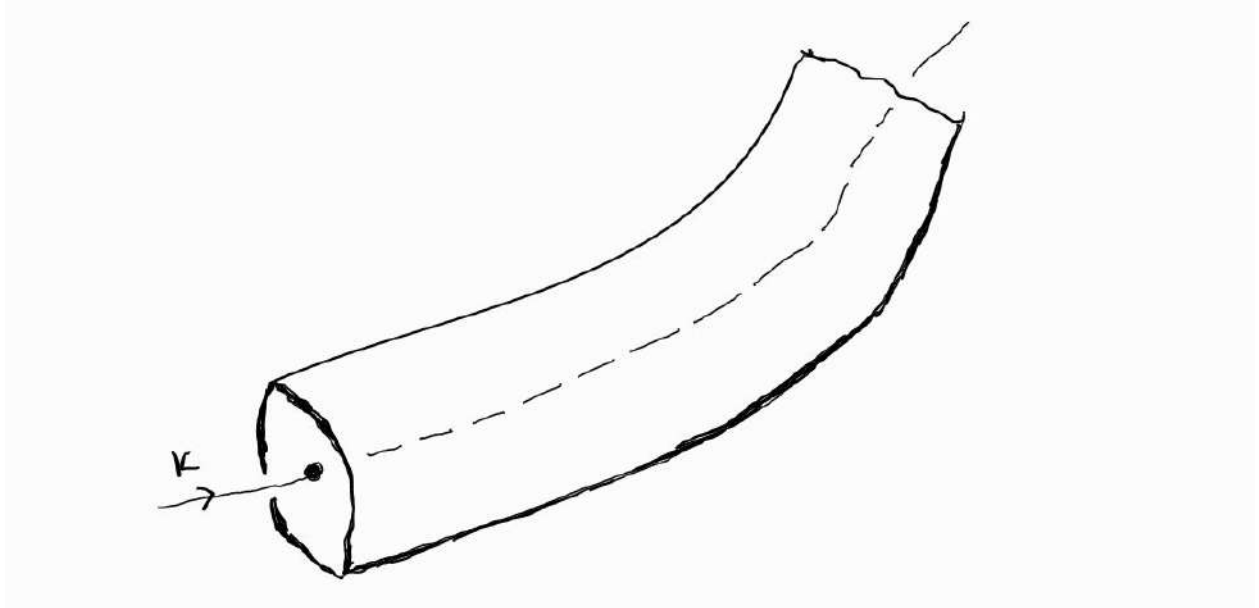
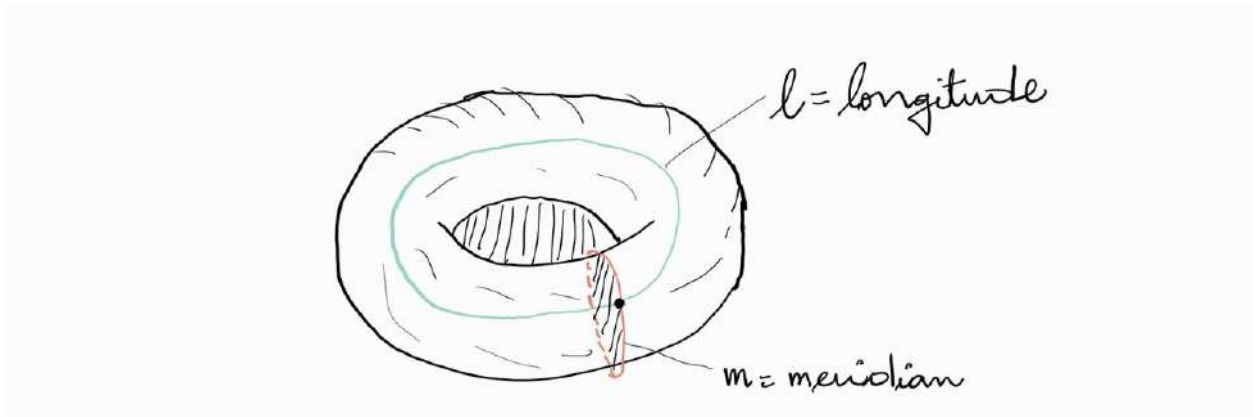


Figure 3.17: A local view of the tubular neighborhood of a knot  $\mathfrak{k}$ .

We can visualize two canonical curves on the boundary of the tubular neighborhood of the knot  $\partial V(\mathfrak{k})$ :

- The longitude  $l$ , which is the boundary of a Seifert surface of the knot.
- The meridian  $m$ , which is the boundary of a disk embedded inside the tubular neighborhood  $V(\mathfrak{k})$

Figure 3.18: Representation of the meridian  $m$  and longitude  $l$ .

### 3.5.3 Geometric structure of the complement of the knot.

**Definition 3.5.4.** If we denote  $V$  as the tubular neighborhood of the knot  $\mathfrak{k}$ , we define  $C = S^3 \setminus V(\mathfrak{k})$  as the complement of the knot  $\mathfrak{k}$ .

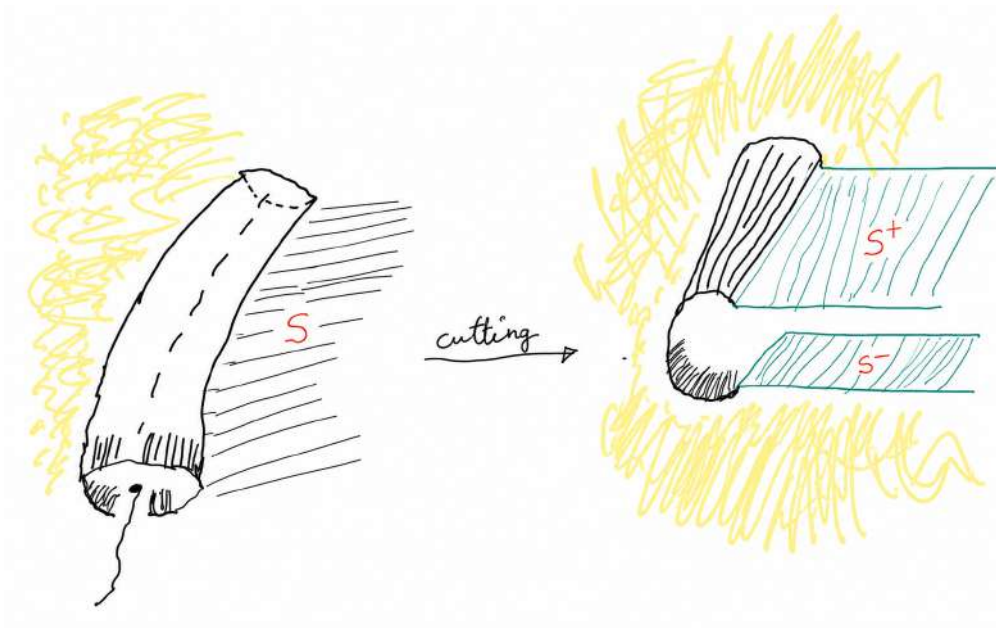
**3.5.5 Cutting the complement of the knot along a Seifert surface.** Let  $\mathfrak{k}$  be an oriented knot and  $S$  a Seifert surface associated with the knot  $\mathfrak{k}$ . Since the Seifert surface  $S$  is embedded and orientable in  $C$ , it has a regular neighborhood that is homeomorphic to  $S \times [-1, 1]$ .

By removing the surface  $S$ , this neighborhood decomposes into two connected components:  $S \times [1, 0)$  and  $S \times (0, -1]$ . Decomposing the complement of the knot along the Seifert surface consists of removing the interior of this regular neighborhood: We thus obtain a new 3-topological manifold.

$$C^* = C \setminus \text{Int}(S \times [-1, 1])$$

whose boundary is  $\partial C^* = S^+ \cup A \cup S^-$ , where  $A \cong S^1 \times I$  represents a sufficiently twisted cylinder.

We say that  $C^*$  is the manifold obtained by cutting out the complement of the knot  $C$  along the Seifert surface  $S$ . There exists a canonical quotient map  $p : C^* \rightarrow C$  that rejoins the two copies  $S^+$  and  $S^-$  to reconstruct the original surface  $S$ : In other words, the quotient map  $p$  allows us to recover the space  $C$  from  $C^*$ .

Figure 3.19: Local representation of  $C^*$ .

This construction plays a crucial role in the study of the infinite cyclic covering of knot complements.

### 3.5.6 Homological property of the complement of the knot.

**Theorem 3.5.7** ([5, Theorem 3.1]). *For any knot  $\mathfrak{k}$ , we have*

$$H_1(C) \cong \mathbb{Z}$$

*and is generated by a meridian  $m_{\mathfrak{k}}$  of the knot  $\mathfrak{k}$ .*

*Proof.* First, we recall that

$$H_n(S^3) = \begin{cases} \mathbb{Z} & \text{if } n = 0, 3 \\ 0 & \text{otherwise} \end{cases}$$

We know that  $\partial V(\mathfrak{k})$  is the torus that is homeomorphic to the product  $S^1 \times S^1$ . Furthermore, since  $S^1$  is compact, by Tychonoff's theorem<sup>7</sup> that the product  $S^1 \times S^1$  is compact. We use Künneth's formula<sup>8</sup> to evaluate the homology groups of  $\partial V(\mathfrak{k})$ , we have:

---

<sup>7</sup>If  $(X_j)_{j \in J}$  denotes a family of topological spaces. The product  $\prod_{j \in J} X_j$  equipped with the product topology is compact if and only if each space  $X_j$  is compact.

<sup>8</sup>If  $X$  and  $Y$  are two compact topological manifolds, there exists an isomorphism (of homology groups)

$$H_n(X \times Y; \mathbb{Z}) \cong \bigoplus_{p+q=n} H_p(X; \mathbb{Z}) \otimes_{\mathbb{Z}} H_q(Y; \mathbb{Z})$$

$$\begin{aligned}
H_0(\partial V(\mathfrak{k})) &\cong H_0(S^1) \otimes_{\mathbb{Z}} H_0(S^1) \\
&\cong \mathbb{Z} \otimes \mathbb{Z} \cong \mathbb{Z}. \\
H_1(\partial V(\mathfrak{k})) &= H_1(S^1 \times S^1) \\
&\cong (H_0(S^1) \otimes_{\mathbb{Z}} H_1(S^1)) \oplus (H_1(S^1) \otimes_{\mathbb{Z}} H_0(S^1)) \\
&\cong (\mathbb{Z} \otimes \mathbb{Z}) \oplus (\mathbb{Z} \otimes \mathbb{Z}) \\
&\cong \mathbb{Z} \oplus \mathbb{Z} \cong \mathbb{Z}^2. \\
H_2(\partial V(\mathfrak{k})) &\cong (H_0(S^1) \otimes H_2(S^1)) \oplus (H_1(S^1) \otimes H_1(S^1)) \oplus (H_2(S^1) \otimes H_0(S^1)) \\
&\cong \underbrace{(\mathbb{Z} \otimes 0)}_{\cong 0} \oplus \underbrace{(\mathbb{Z} \otimes \mathbb{Z})}_{\cong \mathbb{Z}} \oplus \underbrace{(0 \otimes \mathbb{Z})}_{\cong 0} \\
&\cong 0 \oplus \mathbb{Z} \oplus 0 \cong \mathbb{Z}. \\
H_i(\partial V(\mathfrak{k})) &= 0, \quad \forall i > 2 = \dim_{\mathbb{R}} \partial V(\mathfrak{k}). \quad (\text{since } \partial V(\mathfrak{k}) \text{ is a surface.})
\end{aligned}$$

Finally, we obtain:

$$H_n(\partial V(\mathfrak{k})) \cong \begin{cases} \mathbb{Z} & \text{if } n = 0, 2 \\ \mathbb{Z}^2 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Note also that  $D^2$  is contractile, i.e.,  $D^2 \simeq \{*\}$ , so

$$H_i(V(\mathfrak{k})) \cong H_i(S^1 \times D^2) \cong H_i(S^1 \times \{*\}) \cong H_i(S^1) = \begin{cases} \mathbb{Z} & \text{if } i = 0, 1 \\ 0 & \text{else} \end{cases}$$

With all the ingredients in place, we can now evaluate  $H_1(C)$ . To do so, let us consider  $S^3 = V(\mathfrak{k}) \cup C$  and  $V(\mathfrak{k}) \cap C = \partial V(\mathfrak{k})$ .

By applying the Mayer–Vietoris sequence to the pair  $(V(\mathfrak{k}), C)$ , we obtain:

$$\begin{array}{ccccccc}
0 \cdots & \longrightarrow & H_3(\partial V(\mathfrak{k})) & \xrightarrow{i_{\#}^3} & H_3(V(\mathfrak{k})) \oplus_{\mathbb{Z}} H_3(C) & \xrightarrow{j_{\#}^3} & H_3(S^3) \\
& & & & \searrow \partial_{\#}^3 & & \\
& & \nearrow & H_2(\partial V(\mathfrak{k})) & \xrightarrow{i_{\#}^2} & H_2(V(\mathfrak{k})) \oplus_{\mathbb{Z}} H_2(C) & \xrightarrow{j_{\#}^2} & H_2(S^3) \\
& & & & \searrow \partial_{\#}^2 & & \\
& & \nearrow & H_1(\partial V(\mathfrak{k})) & \xrightarrow{i_{\#}^1} & H_1(V(\mathfrak{k})) \oplus_{\mathbb{Z}} H_1(C) & \xrightarrow{j_{\#}^1} & H_1(S^3) \\
& & & & \searrow \partial_{\#}^1 & & \\
& & \nearrow & H_0(\partial V(\mathfrak{k})) & \xrightarrow{i_{\#}^0} & H_0(V(\mathfrak{k})) \oplus_{\mathbb{Z}} H_0(C) & \xrightarrow{j_{\#}^0} & H_0(S^3) \longrightarrow 0
\end{array}$$

By substituting:

$$\begin{array}{ccccccc}
\cdots & \longrightarrow & 0 & \xrightarrow{i_{\#}^3} & H_3(C) & \xrightarrow{j_{\#}^3} & \mathbb{Z} \\
& & & & \searrow \partial_{\#}^3 & & \nearrow \\
& & \mathbb{Z} & \xrightarrow{i_{\#}^2} & H_2(C) & \xrightarrow{j_{\#}^2} & 0 \\
& & & & \searrow \partial_{\#}^2 & & \nearrow \\
& & \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{i_{\#}^1} & \mathbb{Z} \oplus H_1(C) & \xrightarrow{j_{\#}^1} & 0 \\
& & & & \searrow \partial_{\#}^1 & & \nearrow \\
& & \mathbb{Z} & \xrightarrow{i_{\#}^0} & \mathbb{Z} \oplus \mathbb{Z} & \xrightarrow{j_{\#}^0} & \mathbb{Z} \longrightarrow 0
\end{array}$$

The exactitude of the following portion:

$$0 \xrightarrow{\partial_{\#}^2} \mathbb{Z} \oplus \mathbb{Z} \xrightarrow{i_{\#}^1} \mathbb{Z} \oplus H_1(C) \xrightarrow{j_{\#}^1} 0$$

implies that  $i_{\#}^1$  is an isomorphism, so it follows that  $H_1(C) = \mathbb{Z}$ .

We also have  $H_0(C) = \mathbb{Z}$  since the space  $C = \overline{S^3} \setminus V(\mathfrak{k})$  is connected.

□

**Remark 3.5.8.** By using the Mayer-Vietoris exact sequence and an argument regarding the dimension of the space  $C$ , it is possible to show that  $H_n(C) = 0$  for all  $n > 1$ , which we unfortunately do not do in the scope of this thesis. Knowing  $H_1(C)$  will allow us to obtain more information about the Hurewicz morphism, which will be of great use to us in characterizing the elements of the commutator subgroup of the knot group.

Since  $H_1(C) \cong \mathbb{Z}$  implies, by Hurewicz's theorem 2.1.12, that the abelianization of the fundamental group of the complement of the knot  $Ab(\pi_1(C))$  is isomorphic to  $\mathbb{Z}$ , that is

$$\pi_1(C)/[\pi_1(C), \pi_1(C)] \cong \mathbb{Z}.$$

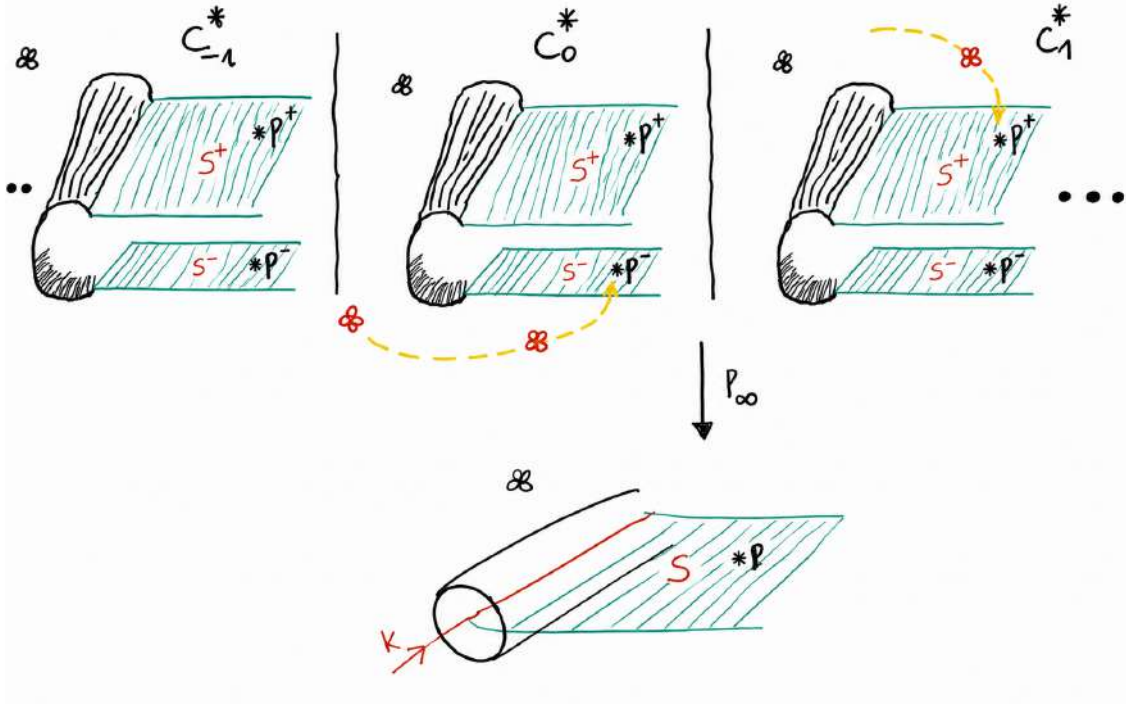
Since the commutator subgroup is normal by definition, by the theorem 2.2.6 it is naturally associated with a regular covering that has an action of  $\mathbb{Z}$  on itself. Since this holds for any knot  $\mathfrak{k}$ , we can study the commutator subgroup of the knot group for any knot. This covering, called the *infinite cyclic covering* of the complement of the knot whose fundamental group is the commutator subgroup of the knot group, will be the central object we will develop in the next chapter.

## 4. Construction of the infinite cyclic cover of the complement of a knot and explicit relationship to commutator subgroup

Many of the proofs in this chapter follow Burdes-Zieschang's book [5], Chapter 4. The infinite cyclic covering of the complement of a knot is a space with a very rich structure but one that is difficult to visualize. Already, the complement of a knot in  $S^3$  is not an easy space to represent geometrically; the situation quickly becomes more complex when considering its associated infinite cyclic covering. Thus, it is necessary to employ a more advanced form of imagination. However, this is not a purely visual intuition: this imagination must be guided by precise logic, as geometric representations quickly become insufficient to fully grasp the construction. To provide a better intuition of the space, the following detailed construction will be accompanied by schematic diagrams.

### 4.1 Construction of the infinite cyclic covering

**Intuitive idea:** Let  $\mathfrak{k}$  be a knot and  $V$  a regular tubular neighborhood of  $\mathfrak{k}$ . Let  $C = S^3 \setminus V(\mathfrak{k})$  denote the complement of the knot. Let  $S$  be a Seifert surface for  $\mathfrak{k}$ . We construct a manifold  $C^*$  with boundary  $\partial C^* = S^+ \cup A \cup S^-$ , by cutting  $C$  along the surface  $S$ . We know that after the cut, every point  $P$  on the Seifert surface  $S$  is duplicated in  $C^*$ , and that all other points in the complement of the surface remain unchanged. Next, we make several (abstract) copies of  $C^*$ . The final step consists of joining  $S_j^-$  with  $S_{j+1}^+$ , i.e.,  $S_j^- \equiv S_{j+1}^+$  for any  $j$ . Concretely, this identification means that if we imagine an object moving in the direction of  $S_j^-$  (intersecting it at point  $P_j^-$ ), it can exit at the next copy  $S_{j+1}^+$  at point  $P_{j+1}^+$ , and we can do this as many times as we like in such a way that we never return to the starting point. The geometric situation can be visualized as in Figure 4.1.

Figure 4.1: Infinite cyclic covering  $C_\infty$ .

We now describe the construction in a more rigorous setting. Since  $S^+$  and  $S^-$  are copies of the Seifert surface  $S$ , there exists a canonical homeomorphism  $r : S^+ \rightarrow S^-$  that identifies these two copies. Now consider a sequence of copies (abstract or otherwise)  $\{C_j^*\}_{j \in \mathbb{Z}}$  of  $C^*$ , along with homeomorphisms  $h_j : C^* \rightarrow C_j^*$ , where  $h_j(x)$  represents the  $j$ th copy of  $x$  on  $C_j^*$  with  $x$  in  $C^*$ .

We define the space  $C_\infty$  as follows:

$$C_\infty = \bigcup_{j=-\infty}^{+\infty} C_j^* / \sim$$

where  $h_j(x) \sim h_{j+1}(r(x))$  for all  $x \in S^+$ : This means that in  $C_\infty$ , we identify  $S^+$  in  $C_j^*$  with  $S^-$  in  $C_{j+1}^*$ .

We define the following correspondence:

$$p_\infty : C_\infty \rightarrow C, \quad \text{by defining } p_\infty(x) = h_j^{-1}(x) \text{ if } x \in C_j^*.$$

**Proposition 4.1.1.** The map  $p_\infty$  is a covering of  $C$ .

*Proof.* For a rigorous proof of Proposition 4.1.1, we refer the reader to [19]. Reproducing the full argument would require a categorical treatment of the construction via pushouts and colimits, which would take us beyond the intended scope of this dissertation.  $\square$

### 4.1.2 Covering Structure of the Infinite Cyclic Covering.

**Proposition 4.1.3.** (4.1 Proposition and Definition chapter 4 Burdes)

The map  $p_\infty : C_\infty \rightarrow C$  is a covering<sup>1</sup> of  $C$  whose covering transformation group is isomorphic to  $\mathbb{Z}$ , i.e.,  $\text{Aut}(p_\infty) = \mathbb{Z}$  (up to isomorphism).

*Proof.* To do this, it suffices to show that the group of automorphisms  $\text{Aut}(p_\infty)$  acts transitively on the fibers  $p_\infty^{-1}(x)$ , for all  $x$  in  $C$ . Let us first prove that  $\text{Aut}(p_\infty) \simeq \mathbb{Z}$ .

Using the notation above, consider  $\tau \in \text{Aut}(p_\infty)$  defined by  $\tau|_{C_j^*} = h_{j+1} \circ h_j^{-1}$ . It is clear that  $\tau$  is a homeomorphism (since it is composed of homeomorphisms  $h_{j+1}$  and  $h_j^{-1}$ ). Furthermore,  $p_\infty \circ \tau = p_\infty$ . Indeed, let  $x \in C_j^*$ , then we have:

$$\begin{aligned} (p_\infty \circ \tau)(x) &= p_\infty(\tau(x)) \\ &= p_\infty(h_{j+1}(h_j^{-1}(x))) \\ &= h_{j+1}^{-1}(h_{j+1}(h_j^{-1}(x))) \\ &= h_j^{-1}(x) \\ &= p_\infty(x). \end{aligned}$$

Thus, it follows that  $\tau$  is a transformation of the covering  $p_\infty$ , i.e.,  $\tau \in \text{Aut}(p_\infty)$ .

**Remark 4.1.4.** For all  $j, m \in \mathbb{Z}$ , we have  $\tau^m(C_j^*) = C_{j+m}^*$ .

Now let's consider the map  $\Phi : \mathbb{Z} \rightarrow \text{Aut}(p_\infty)$ , where  $\Phi(k) = \tau^k$ . We will verify that  $\Phi$  is an isomorphism (of groups).

- It is clear that  $\Phi$  is well-defined.
- $\Phi$  is a group morphism. Indeed, for  $j$  and  $m$  relative integers:  $\Phi(m+n) = \tau^{m+n} = \tau^m \circ \tau^n = \Phi(m) \circ \Phi(n)$  and  $\Phi(0) = \tau^0 = \text{id}_{C_\infty}$ .
- Let  $m$  and  $n$  be two relative integers such that  $\Phi(m) = \Phi(n)$ . We want to show that  $m = n$ , so we have:

$$\begin{aligned} \Phi(m) = \Phi(n) &\implies \tau^m = \tau^n \\ &\implies \tau^m(C_0^*) = \tau^n(C_0^*) \\ &\implies C_m^* = C_n^* \\ &\implies m = n \end{aligned}$$

The last equality follows from the fact that in the representation of  $C_\infty$ , the  $C_j^*$  are uniquely determined: it follows that  $\Phi$  is injective.

---

<sup>1</sup>Note:  $p_\infty$  is a covering on  $C$  and not on  $C^*$ , but we use copies of  $C^*$  to construct this covering.



- Let  $\gamma \in \text{Aut}(p_\infty)$ . Find an integer  $m$  such that  $\Phi(m) = \gamma$ . We have:

$$\begin{aligned}
 \Phi(m) = \gamma &\implies \tau^m = \gamma \quad (\text{by definition of } \Phi) \\
 &\implies \tau^m(C_j^*) = \gamma(C_j^*) \quad (\text{we can set } \gamma(C_j^*) = C_{n_{j,\gamma}}) \\
 &\implies C_{m+j}^* = C_{n_{j,\gamma}} \\
 &\implies m + j = n_{j,\gamma} \\
 &\implies m = n_{j,\gamma} - j
 \end{aligned}$$

It follows that  $\Phi$  is surjective.

**Conclusion:** We have shown that  $\Phi$  is a bijection and that  $\Phi \in \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \text{Aut}(p_\infty))$ ; it follows, therefore, that  $\mathbb{Z}$  is isomorphic to  $\text{Aut}(p_\infty)$ . And it is clear that  $\tau$  is a generator of the group  $\text{Aut}(p_\infty)$ .

Now we need only show that  $p_\infty : C_\infty \rightarrow C$  is a Galois covering, i.e., that  $\text{Aut}(p_\infty)$  acts transitively on each fiber  $p_\infty^{-1}(x)$ , with  $x \in C$ . To do this, fix a point  $x$  in  $C$  and consider the following correspondence:

$$\phi : \text{Aut}(p_\infty) \times p_\infty^{-1}(x) \longrightarrow p_\infty^{-1}(x), \quad \phi(\gamma, \tilde{x}) = \gamma(\tilde{x}) = \tau^n(\tilde{x}).$$

We will verify that  $\phi$  is a transitive action of the group  $(\text{Aut}(p_\infty), \circ, \text{id}_{C_\infty})$  on the space  $p_\infty^{-1}(x)$ .

- The equality  $p_\infty \circ \tau = p_\infty$  implies  $p_\infty \circ \tau^n = p_\infty$  for any positive integer  $n$ . Thus, for  $n \in \mathbb{Z}$  and  $\tilde{x} \in p_\infty^{-1}(x)$ , we have  $p_\infty(\tau^n(\tilde{x})) = p_\infty(\tilde{x}) = x$ , so  $\tau^n(\tilde{x}) = \phi(\tau^n, \tilde{x}) \in p_\infty^{-1}(x)$ : thus  $\phi$  is well-defined.
- Let  $m, n \in \mathbb{Z}$  and  $\tilde{x} \in p_\infty^{-1}(x)$ , then we have:

$$\begin{aligned}
 \phi(\tau^n, \phi(\tau^m, \tilde{x})) &= \phi(\tau^n, \tau^m(\tilde{x})) \\
 &= \tau^n(\tau^m(\tilde{x})) \\
 &= \tau^{n+m}(\tilde{x}) \\
 &= \phi(\tau^{n+m}, \tilde{x}) \\
 &= \phi(\tau^n \circ \tau^m, \tilde{x})
 \end{aligned}$$

Furthermore, we have  $\phi(\text{id}_{C_\infty}, \tilde{x}) = \text{id}_{C_\infty}(\tilde{x}) = \tilde{x}$ .

So  $\phi$  is a group action. As for the transitivity of the action  $\phi$ , let  $\tilde{x}$  and  $\tilde{y}$  be two (distinct) elements of  $p_\infty^{-1}(x)$ . Let us find a relative integer  $k$  such that  $\phi(\tau^k, \tilde{x}) = \tilde{y}$ , i.e.,  $\tau^k(\tilde{x}) = \tilde{y}$ . There exist integers  $i$  and  $j$  such that  $\tilde{y} \in C_i^*$  and  $\tilde{x} \in C_j^*$ . The condition  $\tau^k(\tilde{x}) = \tilde{y}$  is equivalent to  $\tau^k(C_j^*) = C_i^*$ , which implies  $k + j = i$ , i.e.,  $k = i - j$ .

□

**4.1.5 Conclusion.** The construction above describes the infinite cyclic covering as an infinite union of copies of the complement of the knot cut along the Seifert surface, arranged according to the action of the group  $\mathbb{Z}$ . Since the group of transformations of the covering is isomorphic to  $\mathbb{Z}$ , this is why it is called an infinite cyclic covering. This construction can also be formulated more abstractly using pushouts, but here we prefer a geometric approach better suited to the framework of knot theory [19]. This covering plays a fundamental role in the study of the commutator subgroup of the knot group.

## 4.2 The structure theorem for the commutator subgroup

The second part of this chapter is devoted to the structure of the commutator subgroup. We begin by recalling Neuwirth's lemma, which will be of great use to us in proving the structure theorem for the commutator subgroup.

**Lemma 4.2.1** (Neuwirth, [5, Lemma 4.5]). *When  $S$  is a Seifert surface of minimal genus spanning the knot  $\mathfrak{k}$  the inclusions  $i^\pm : S^\pm \rightarrow C^*$  induce monomorphisms  $i_\#^\pm : \pi_1 S^\pm \rightarrow \pi_1 C^*$ .*

*Proof.* Suppose that  $i_\#^+$  is non-injective, i.e.,  $\text{Ker}(i_\#^+) \neq \{1\}$ . There exists (at least) one loop  $\eta$  in  $\pi_1 S^+ \setminus \{1\}$  such that  $i_\#^+([\eta]) = 1$  (in particular,  $\eta \neq 1$  in  $S^+$ ). By applying the loop theorem<sup>2</sup>, we can find a disk  $\delta \subset C^*$  such that  $\partial\delta = \eta = \delta \cap \partial C = \delta \cap S^+$ . The next step is to perform a surgery on  $S^+$ : We obtain a new copy<sup>3</sup> of a Seifert surface:

$$S_1^+ = (S^+ \setminus U(\delta)) \cup \delta_1 \cup \delta_{-1},$$

where  $U(\delta) = [-1, 1] \times \delta$  is a regular neighborhood of  $\delta$  in  $C$ , and  $\delta_i = \{i\} \times \delta$ .

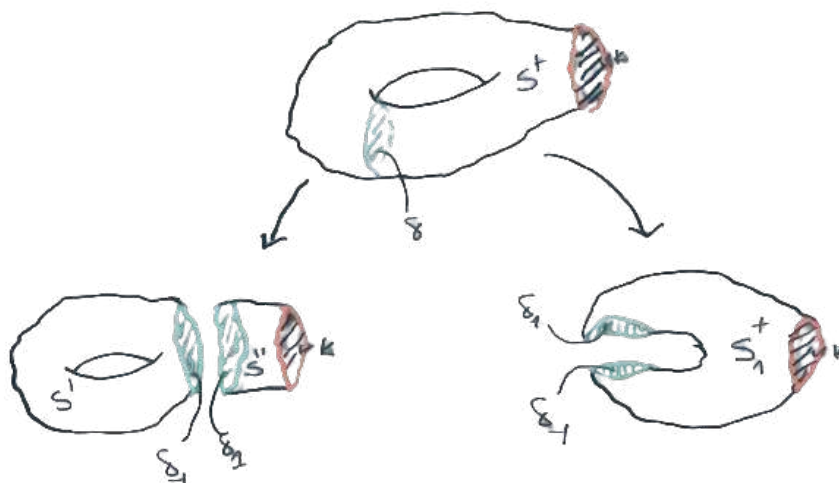


Figure 4.2: surgery of  $S$

<sup>2</sup>Put this theorem in the Appendix

<sup>3</sup>In fact, it is possible to construct a Seifert surface  $S_1$  from  $S_1^+$ .

We distinguish between two situations, as shown in Figure 4.2.

**Case 1:**  $S_1^+$  is connected.

In this case,  $g(S_1) = g(S_1^+) = g(S^+) - 1 < g(S^+) = g(S)$ : this contradicts the minimality of  $g(k)$ .

**Case 2:**  $S_1^+$  is not connected.

Here,  $S_1^+$  decomposes into (at least) two connected components  $S_1^+ = S' \sqcup S''$ . Since the construction consists of adding a regular neighborhood of a disk whose boundary lies in  $S^+$ , one of the components of  $S_1^+$  has the knot  $k$  as its boundary.

Let  $S'$  be this component: then  $S'$  is a Seifert surface for the knot  $k$ . Furthermore, the surgery performed along  $\delta$  strictly reduces the topological complexity: indeed, we have  $g(S') < g(S^+)$ : This again contradicts the minimality of  $g(k)$ .

**Conclusion:** In all cases, we obtain a contradiction. Therefore,  $i_{\#}^+$  is a monomorphism. We proceed in the same way to show that  $i_{\#}^-$  is a monomorphism.

□

#### Theorem 4.2.2. (Structure of the commutator subgroups)

We have the following situations:

- (a) If the commutator subgroup  $\mathcal{G}'$  of a knot group  $\mathcal{G}$  is finitely generated, then  $\mathcal{G}'$  is a free group of rank  $2g$  where  $g$  is the genus of the knot. In fact,  $\mathcal{G}' = \pi_1(S)$ ,  $S$  a Seifert surface of genus  $g$ .
- (b) If  $\mathcal{G}'$  cannot be finitely generated, then

$$\mathcal{G}' = \cdots \mathcal{A}_{-1} *_{\mathcal{B}_{-1}} \mathcal{A}_0 *_{\mathcal{B}_0} \mathcal{A}_1 *_{\mathcal{B}_1} \mathcal{A}_2 \cdots$$

and the generator  $t$  of the group of covering transformations of  $p_{\infty} : C_{\infty} \rightarrow C$  induces an automorphism  $\tau$  of  $\mathcal{G}'$  such that  $\tau(\mathcal{A}_j) = \mathcal{A}_{j+1}$ ,  $\tau(\mathcal{B}_j) = \mathcal{B}_{j+1}$ . Here  $\mathcal{A}_j \cong \pi_1 C^*$ ,  $\mathcal{B}_j \cong \pi_1 S \cong \mathcal{F}_{2g}$  and  $\mathcal{B}_j$  is a proper subgroup of  $\mathcal{A}_j$  and  $\mathcal{A}_{j+1}$ . (The subgroups  $\mathcal{B}_j$  and  $\mathcal{B}_{j+1}$  do not coincide.)

*Proof.* In what follows, we will limit ourselves to proving the first part (a) of this theorem. Indeed, the second part does not appear later in this thesis, and its proof requires more sophisticated tools, notably the Brown–Crowell lemma ([5], Lemma 4.7).

We start with the complement of a knot  $k$ , denoted  $C$ . We apply the construction 3.5.5 to  $C$ , i.e., we cut  $C$  along a Seifert surface  $S$  of minimal genus of the knot  $k$ : We obtain a topological 3-manifold  $C^*$ .

Consider the infinite cyclic covering  $p_{\infty} : C_{\infty} \rightarrow C$

$$C_\infty = \bigcup_{j=-\infty}^{+\infty} C_j^*, \text{ where } C_j^* \cap C_{j+1}^* = S_j, \quad \forall j \in \mathbb{Z}.$$

Let  $Y_0 = C_0^*$ . For each  $n \geq 1$ , define

$$Y_n = \bigcup_{j=0}^n C_j^*.$$

Then  $Y_0 \subseteq Y_1 \subseteq Y_2 \subseteq \dots$  and  $C_\infty = \bigcup_{n \geq 0} Y_n$ . Set  $\beta_n = \pi_1(Y_n)$ . Applying the Seifert–Van Kampen theorem inductively, we obtain

$$\beta_{n+1} = \beta_n *_{\pi_1(S_n)} \pi_1(C_{n+1}^*), \quad n \geq 0.$$

Hence,

$$\beta_0 \subseteq \beta_1 \subseteq \beta_2 \subseteq \dots$$

Since  $C_\infty = \bigcup_{n \geq 0} Y_n$ , it follows that

$$\mathfrak{G}' = \pi_1(C_\infty) = \varinjlim \beta_n = \bigcup_{n \geq 0} \beta_n.$$

By assumption,  $\mathfrak{G}'$  is finitely generated, which means there exist elements  $g_1, \dots, g_k$  such that  $\mathfrak{G}' = \langle g_1, \dots, g_k \rangle$ . Furthermore, given that  $\mathfrak{G}' = \bigcup_n \beta_n$ , for each  $i$  there exists  $n_i$  such that  $g_i \in \beta_{n_i}$ .

Now, let  $m = \max(n_1, \dots, n_k)$ . It follows that  $\mathfrak{G}' = \beta_m$  (1) (indeed,  $\beta_m \subset \bigcup_n \beta_n = \mathfrak{G}'$  and conversely, for all  $i$ ,  $g_{n_i} \in \beta_{n_i} \subset \beta_m$  since the sequence  $(\beta_n)_n$  is increasing, so  $\mathfrak{G}' \subset \beta_m$ ). Equality (1) implies that starting from a certain rank  $m$ , we have  $\beta_{m+j} = \beta_m$  for all  $j \geq 1$ . In particular, this leads to the following result:

$$\beta_m *_{\pi_1(S_m)} \pi_1(C_{m+1}^*) = \beta_{m+1} = \beta_m \implies \pi_1(S) \cong \pi_1(S_m) = \pi_1(C_{m+1}^*) \cong \pi_1(C^*)$$

However, we proved in Chapter 3, in the section concerning the fundamental group of a Seifert surface, that  $\pi_1(S) \cong \mathbb{F}_{2g}$  therefore  $\pi_1(C^*) \cong \mathbb{F}_{2g}$ .  $\square$

### 4.2.3 A Strange Family of Knots: Three-Strand Pretzel Knots with Odd Parameters.

This subsection follows from Exercise 4.10 in [5]; we study three-strand pretzel knots of the form  $\mathfrak{p}(2p+1, 2q+1, 2r+1)$ ,  $p, q, r \in \mathbb{Z}$ . Our goal is to compute the induced morphisms  $i_{\#}^{\pm} : \pi_1(S) \rightarrow \pi_1(C^*)$  and  $i_*^{\pm} : H_1(S) \rightarrow H_1(C^*)$ , where  $S$  is a Seifert surface of minimal genus bordering the knot under consideration. Using these calculations, we will be able to classify 3-strand Pretzel knots with odd parameters in terms of their derived subgroups.

**Definition 4.2.4** ([13], 2 Pretzel knots). A  $(p_1, \dots, p_k)$ -pretzel knot  $\mathfrak{p}(p_1, \dots, p_k)$  is a union of  $k$ -pairs of strands twisted  $p_1, \dots, p_k$  times and attached along the tops and bottoms. The twists are oriented according to whether  $p_i$  is positive or negative.

For a pretzel knot  $\mathfrak{k}$  of type  $\mathfrak{p}(2p+1, 2q+1, 2r+1)$ , the genus of the knot is 1 [[8], Theorem 1]. If we denote by  $S$  the Seifert surface associated with the knot  $\mathfrak{k}$ , then  $S$  is homeomorphic (abstractly) to a torus with a small disk removed. We obtain:

- We have  $H_1(S) \cong \mathbb{Z}^2 = \langle a, b \rangle$  and  $H_1(C^*) \cong \mathbb{Z}^2 = \langle x, y \rangle$ : where  $a, b$  are the generating curves of  $H_1(S)$  and  $x, y$  are the generating curves of  $H_1(C^*)$  such that  $lk(x, a) = 1$  and  $lk(y, b) = 1$ .

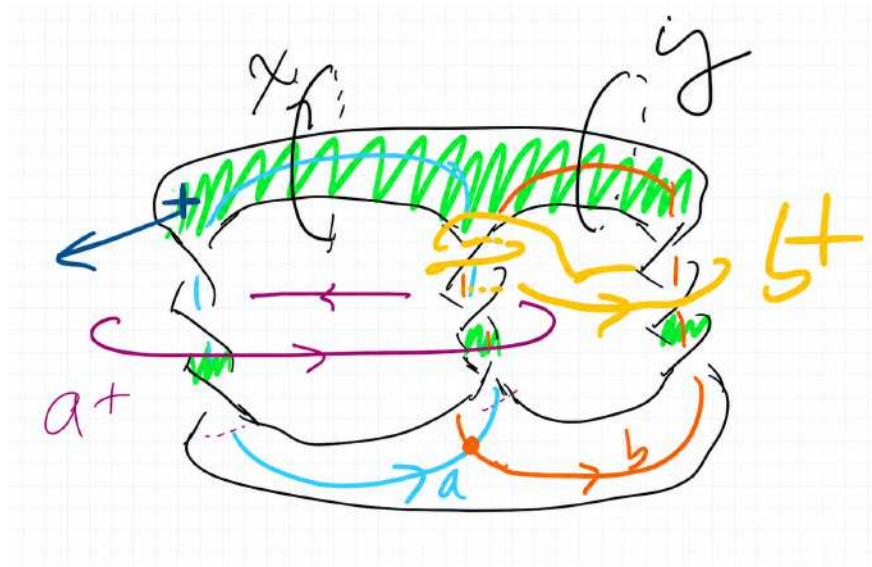


Figure 4.3: Case of knot Pretzel  $\mathfrak{p}(-3, 3, 3)$

Moreover, since  $i_{\#}^{\pm} : H_1(S^{\pm}) \rightarrow H_1(C^*)$  is a morphism<sup>4</sup>, then it is entirely determined by the image of the generators of the basis of  $H_1(S)$ , i.e.,  $i_{\#}^{\pm}(a)$  and  $i_{\#}^{\pm}(b)$ .

If we denote  $a^+$  and  $b^+$  as the respective pushed copies of the generators  $a$  and  $b$ , using Figure 4.3, we can write

$$i_{\#}^{\pm}(a) = a^+ = mx + ny \text{ and } i_{\#}^{\pm}(b) = b^+ = m'x + n'y$$

where  $m$  and  $n$  represent the number of crossings of the curves  $x$  and  $y$ , respectively, with the curve  $a^+$ , and  $m'$  and  $n'$  represent the number of crossings of the curves  $x$  and  $y$ , respectively, with the curve  $b^+$ . We know that  $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}^2, \mathbb{Z}^2) \cong M_{2,2}(\mathbb{Z})$ . Thus, the morphism  $i_{\#}^{\pm}$  can be represented by a  $2 \times 2$  integer matrix. Given the definition of the morphism  $i_{\#}^{\pm}$ , its representative matrix in well-chosen bases  $(\mathcal{B}_{H_1(S)} = \{a, b\})$  and  $\mathcal{C}_{H_1(C^*)} = \{x, y\})$  coincides with the Seifert matrix associated with  $S$ .

▪

$$i_{\#}^+ = \begin{cases} \text{isomorphism if } \mathfrak{G}' \text{ is finitely generated by Theorem 4.2.2,} \\ \text{monomorphism (by Lemma 4.2.1) but not surjective if } \mathfrak{G}' \text{ is not finitely generated.} \end{cases}$$

<sup>4</sup>the morphism  $i_{\#}^{\pm}$  takes a curve described on the Seifert surface and maps it into the space  $C^*$ .

Now we will determine which of the Pretzel knots are finitely generated.

If we assume that  $\mathfrak{G}'$  is finitely generated, this implies that  $i_{\#}^{\pm}$  is an isomorphism, i.e.,  $\pi_1(S) \cong \pi_1(C^*)$ . This implies that  $i_*^{\pm}$  is an isomorphism such that the following diagram is commutative,

$$\begin{array}{ccc} \pi_1(S) & \xrightarrow{i_{\#}^{\pm}} & \pi_1(C^*) \\ p \downarrow & & \downarrow p \\ H_1(S) \cong \text{Ab}(\pi_1(S)) & \xrightarrow{i_*^{\pm}} & \text{Ab}(\pi_1(C^*)) \cong H_1(C^*) \end{array}$$

Therefore, the fact that  $\mathfrak{G}'$  is finitely generated implies that  $i_*^{\pm} : H_1(S) \rightarrow H_1(C^*)$  is an isomorphism. We know that for the pretzel knot  $\mathfrak{p}(p, q, r)$  the associated Seifert matrix [3, Corollary 7.2] is:

$$V_{\mathfrak{k}} = \frac{1}{2} \begin{pmatrix} p+q & q+1 \\ q-1 & q+r \end{pmatrix}.$$

In our case,  $\mathfrak{k} = \mathfrak{p}(2p+1, 2q+1, 2r+1)$ , we get:

$$V_{\mathfrak{k}} = \begin{pmatrix} p+q+1 & q+1 \\ q & q+r+1 \end{pmatrix}.$$

Therefore,  $i_*^{\pm}$  is an isomorphism  $\iff V_{\mathfrak{k}}$  is invertible and  $V_{\mathfrak{k}}^{-1} \in M_2(\mathbb{Z})$ . We have

$$V_{\mathfrak{k}}^{-1} = \frac{1}{\det(V_{\mathfrak{k}})} \text{Com}(V_{\mathfrak{k}})^T \in M_2(\mathbb{Z}).$$

$$V_{\mathfrak{k}}^{-1} \in M_2(\mathbb{Z}) \iff \frac{1}{\det(V_{\mathfrak{k}})} \in \mathbb{Z} \iff \det(V_{\mathfrak{k}}) = \pm 1.$$

Now

$$\begin{aligned} \det(V_{\mathfrak{k}}) &= (p+q+1)(q+r+1) - q(q+1) \\ &= pq + pr + qr + p + q + r + 1 - q^2 - q \\ &= pq + pr + qr + p + q + r + 1. \end{aligned}$$

Therefore:

$$\begin{aligned} \det(V_{\mathfrak{k}}) = \pm 1 &\iff pq + pr + qr + p + q + r + 1 = \pm 1 \\ &\iff (p+1)(q+1)(r+1) - pqr = \pm 1 \\ &\implies \begin{cases} \text{if } |p|, |q|, |r| \geq 2, \text{ there is no solution.} \\ \text{if at least one of the parameters } p, q, r \in \{-1, 0, 1\}, \text{ there is a solution.} \end{cases} \end{aligned}$$

The three-strand Pretzel knots with odd parameters that have a commutator subgroup are those for which one of the three characteristic parameters is in the set  $\{-1, 0, 1\}$ .

**Conclusion:** What we take away from this example is that among the three-strand Pretzel knots with odd parameters, there are very few that have finitely generated commutator subgroups, and we will see in the next chapter that this result allows us to observe that this class of Pretzel knots are the only Pretzel knots that are fibred.

The construction of the infinite cyclic covering thus provides the ideal topological framework for understanding the subgroup of commutators of the knot group. The results of this chapter will serve as our main tools in the proof of Stallings' theorem, to which the next chapter is devoted.

## 5. Statement and proof of Stallings fibration theorem for knot complements

In the previous chapters, we have established the fundamental tools of knot theory necessary for studying the topological properties of knot complements. In particular, we have constructed the infinite cyclic covering. These various results lead to a central question in knot theory: under what conditions does the complement of a knot fibres over the circle  $S^1$ ?

The concept of a fibred knot plays a central role in the theory of three-dimensional manifolds. Intuitively, a knot is said to be fibred when its complement admits a fibration over the circle whose fibers are minimal Seifert surfaces of the knot. This property establishes a direct link between the topology of the knot's complement and the algebraic structure of the fundamental group of the knot's complement. The central result of this chapter is Stallings's fibration theorem, which establishes an algebraic criterion for detecting the existence of such a fibration over the circle  $S^1$ .

The objective of this chapter is threefold. First, we will present the statement of Stallings's theorem in the context of knot complements. Next, we will detail the main ideas involved in its proof within a geometric framework. Finally, we will provide a geometric interpretation of Stallings's fibration theorem using the pretzel knot  $p(1, -1, 3)$  as an example.

### 5.1 Stallings' Fibration Theorem

Stallings' Fibration Theorem, in its general form, states that:

**Theorem 5.1.1** (Stallings Theorem [5, Theorem 5.6, page 71]). *Let  $M$  be a compact irreducible 3-manifold (this means that in  $M$  every 2-sphere bounds a 3-ball). Assume that  $\varphi : \pi_1(M) \rightarrow \mathbb{Z}$  is an epimorphism with a finitely generated kernel. Then:*

- (a)  *$\text{Ker}\varphi$  is isomorphic to the fundamental group of a compact surface  $S$ .*
- (b)  *$M$  can be locally trivially fibred over  $S^1$  with fiber  $S$  if  $\text{Ker}\varphi \not\cong \mathbb{Z}_2$ .*

We can find the complete proof in his original paper [21]. In this paper, we focus on the specific case where the 3-manifold is the complement of a knot, which is a space sophisticated enough to be interesting and simple enough to allow us to provide a complete proof.

#### 5.1.2 Stallings Fibration Theorem for Knot Exteriors.

**Theorem 5.1.3** (Stallings Fibration Theorem [5, Theorem 5.1]+[17, Proposition 3, page 324]). *The complement  $C = \overline{S^3 \setminus V(\mathfrak{k})}$  of a knot  $\mathfrak{k}$  fibers locally trivially over  $S^1$  with Seifert surfaces of genus  $g$  (the genus of the knot) as fibers if and only if the commutator subgroup  $\mathfrak{G}'$  of the knot group is finitely generated, i.e.,  $\mathfrak{G}'$  is isomorphic to  $\mathbb{F}_{2g}$ .*



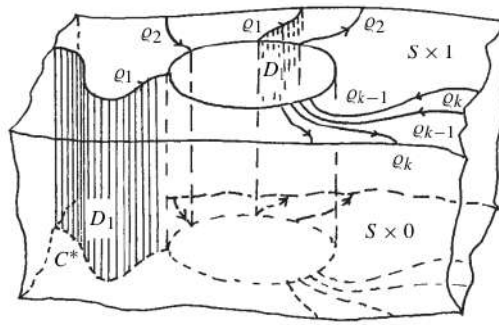


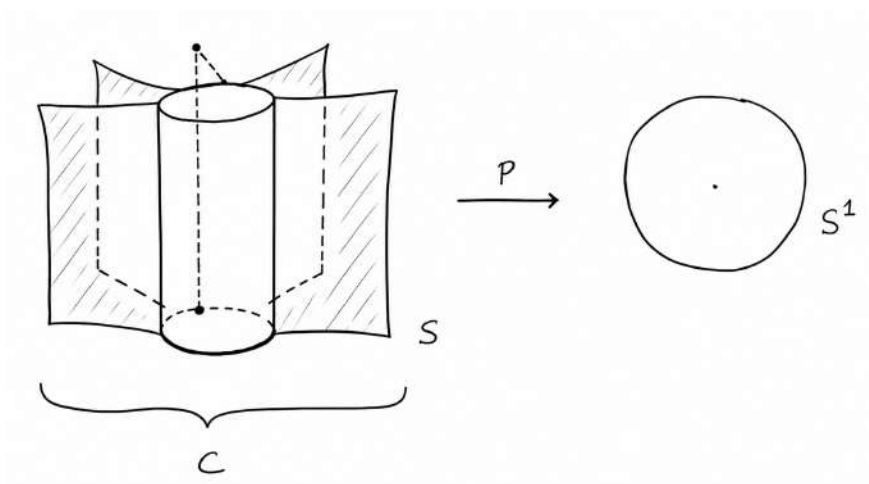
Figure 5.1: Fiber bundle structure of the knot complement

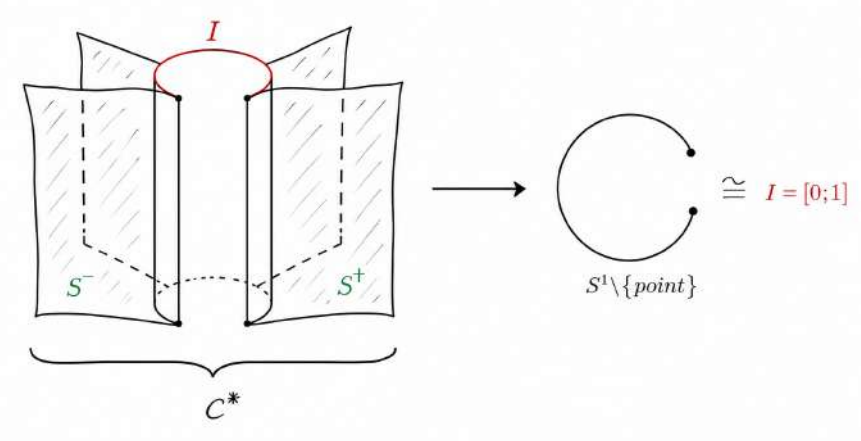
We will proceed with the proof step by step. It relies on several fundamental theorems of three-dimensional geometry, including Alexander's theorem, Dehn's lemma, Nielsen's theorem, and the loop theorem. We have included them in the Appendix along with the corresponding references. Since mapping class groups are not a central topic of this dissertation, we do not develop their theory in detail. We assume familiarity with the basic notions and classical results that will be used in the proof, notably the fact that the mapping class group of the disk is trivial, see the book [6].

## 5.2 Proof of Stallings' fibration theorem for knot exteriors

*Proof.* (From Theorem 5.1.3)

- i) Suppose that  $C$  fibers locally trivially over the circle  $S^1$ . By slicing  $C$  along the Seifert surface as described in paragraph 3.5.5, it follows that  $C^*$  is a trivial fibered space over the interval  $[0; 1]$ , i.e.,  $C^* \cong S \times I$ .


 Figure 5.2: Before cutting of  $C$

Figure 5.3: After cutting of  $C$ 

By using the infinite cyclic covering  $C_\infty$  constructed in Section 4.1, we obtain:

$$C_\infty = \left( \bigcup_{j=-\infty}^{+\infty} C_j^* \right) / \sim \cong \left( \bigcup_{j=-\infty}^{+\infty} (S \times [0; 1])_j \right) / \sim \cong S \times \mathbb{R}$$

Topologiquement on sait que  $\mathbb{R}$  est homéomorphe à l'intervalle unité  $I$ . Par extension de cette homéomorphisme sur le produit  $S \times \mathbb{R}$ , et on montre facilement que  $S \times \mathbb{R}$  est homéomorphe au produit  $S \times I$ , i.e  $S \times \mathbb{R} \cong S \times I$ .

Finalement, on obtient que  $C_\infty \cong S \times I$ , which implies the following isomorphism of fundamental groups:

$$\pi_1(C_\infty) \cong \pi_1(S \times [0; 1]) \cong \pi_1(S) \times \pi_1(I) \cong \pi_1(S) = \mathbb{F}_{2g}$$

However, by definition 3.5.8 of the covering  $C_\infty$ , we know that  $\mathfrak{G}' = \pi_1(C_\infty)$ , so  $\mathfrak{G}' \cong \mathbb{F}_{2g}$ . Consequently, the commutator subgroup  $\mathfrak{G}'$  of the knot group is finitely generated.

- ii) Conversely, suppose that the commutator subgroup  $\mathfrak{G}'$  of the knot group  $\mathfrak{k}$  is finitely generated. It follows from Theorem 4.2.2 that the canonical inclusions  $i_\pm : S^\pm \rightarrow C^*$  induce fundamental group isomorphisms  $i_\#^\pm : \pi_1(S^\pm) \rightarrow \pi_1(C^*)$ . Let  $\gamma$  denote the meridian of the tubular neighborhood  $V(\mathfrak{k})$  based at a point  $P$  on  $\partial S$ . We know that after cutting  $C$  along the surface  $S$ , all points on the Seifert surface split into two: the curve  $\gamma$  (in  $C^*$ ) becomes a path connecting the point  $P^+ = P \times \{1\}$  to the point  $P^- = P \times \{0\}$ . Let  $\omega$  be a loop in  $\pi_1(S^-, P^-)$  and let  $i_\#^-(\omega) = \omega^-$  its copy slightly shifted in  $\pi_1(C^*, P^-)$ . By Lemma 2.1.10, the loop  $\gamma\omega^-\bar{\gamma} = \phi_\gamma(\omega^-)$  is a loop in  $\pi_1(C^*, P^+)$ . Since  $i_\#^+$  is an isomorphism (of fundamental groups), there exists a unique loop  $\omega^+$  in  $\pi_1(S^+, P^+)$  such that  $i_\#^+(\omega^+) = \phi_\gamma(\omega^-) = \gamma\omega^-\bar{\gamma}$  in  $\pi_1(C^*, P^+)$ .

If we set  $g(\omega) = \omega^+$ , this implies that the following diagram

$$\begin{array}{ccc}
\pi_1(S^-, P^-) & \xrightarrow{g} & \pi_1(S^+, P^+) \\
\downarrow i_{\#}^- \cong & & \cong \downarrow i_{\#}^+ \\
\pi_1(C^*, P^-) & \xrightarrow[\cong]{\phi_{\gamma}} & \pi_1(C^*, P^+)
\end{array}$$

commutes, i.e.,  $i_{\#}^+ \circ g = \phi_{\gamma} \circ i_{\#}^-$ .

We consider the following correspondence:

$$f_{\#} : \pi_1(S^-, P^-) \longrightarrow \pi_1(S^+, P^+)$$

$$\omega \longmapsto \omega^+ := \left\{ \begin{array}{l} \text{the element of } \pi_1(S^+, P^+) \text{ such that} \\ i_{\#}^+(\omega^+) = \phi_{\gamma}(\omega^-) \text{ in } \pi_1(C^*, P^+) \text{ and} \\ i_{\#}^-(\omega) = \omega^- \end{array} \right\}$$

We will show that  $f_{\#}$  is an isomorphism. Indeed,

- $f_{\#}$  is well-defined<sup>1</sup> according to the previous construction. Therefore,  $f_{\#}$  is a map.
- It goes without saying that  $f_{\#}$  is a group morphism.
- Let  $\omega_1$  and  $\omega_2$  be two loops in  $\pi_1(S^-, P^-)$  such that  $f_{\#}(\omega_1) = f_{\#}(\omega_2)$ . We want to show that  $\omega_1 = \omega_2$ . By definition of  $f_{\#}$ , we have  $i_{\#}^+(\omega_1^+) = \phi_{\gamma}(\omega_1^-)$  such that  $i_{\#}^-(\omega_1) = \omega_1^-$ , we also have  $i_{\#}^+(\omega_2^+) = \phi_{\gamma}(\omega_2^-)$  such that  $i_{\#}^-(\omega_2) = \omega_2^-$ . From these informations, we can deduce

$$\begin{aligned}
f_{\#}(\omega_1) = f_{\#}(\omega_2) &\implies \omega_1^+ = \omega_2^+ \quad (\text{from definition of } f_{\#}) \\
&\implies i_{\#}^+(\omega_1^+) = i_{\#}^+(\omega_2^+) \\
&\implies \phi_{\gamma}(\omega_1^-) = \phi_{\gamma}(\omega_2^-) \\
&\implies \omega_1^- = \omega_2^- \quad (\text{because } \phi_{\gamma} \text{ is injective}) \\
&\implies i_{\#}^-(\omega_1) = i_{\#}^-(\omega_2) \\
&\implies \omega_1 = \omega_2 \quad (\text{because } i_{\#}^- \text{ is injective})
\end{aligned}$$

This concludes that the morphism  $f_{\#}$  is injective.

- $f_{\#}$  is surjective. Indeed, if we take an arbitrary loop  $\eta$  in  $\pi_1(S^+, P^+)$ , starting the previous construction in the reverse direction leads to the result.

We obtain that  $f_{\#}$  is a bijective morphism, so it is a group isomorphism.

By Nielsen's theorem 5.3.2, there exists a homeomorphism  $f : S^- \longrightarrow S^+$  that induces  $f_{\#}$ .

---

<sup>1</sup>**Note:** One must not confuse  $\omega^-$  and  $\omega^+$  with the images on  $S^-$  and  $S^+$  of an initial loop  $\omega$  that might exist in  $C^*$

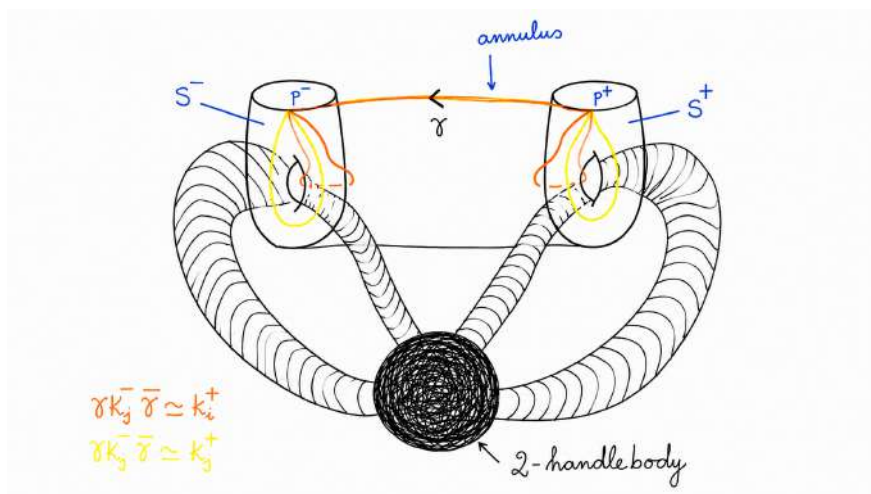


Figure 5.4: Geometric representation of the situation

Now let  $\{k_i \mid 1 \leq i \leq 2g\}$  be a canonical system of generating curves for the minimal Seifert surface  $S^-$ . By linear extension of  $f_\#$  to the generators of the basis of  $\pi_1(S^-, P^-)$  (which are none other than the  $k_i$ ), we have  $f_\#(k_i) = k_i^+$  and the condition  $i_\#^+(k_i^+) = \phi_\gamma(k_i^-)$  implies that

$$\gamma \cdot k_i^- \cdot \bar{\gamma} \cdot (i_\#^+(k_i^+))^{-1} = \rho_i' \simeq 0 \text{ in } \pi_1(C^*, P^+)$$

Furthermore, looking at Figure 5.4, we can see that the curves  $\rho_i$  coincide along the path  $\gamma$ . If we consider a neighborhood (we can take the disk  $D^2$ ) of each point on  $\gamma$ , we obtain a finite cylinder  $N(\gamma) \cong D^2 \times I$ , which can be viewed as the regular neighborhood of the path  $\gamma$ : The underlying idea is that the curves  $\rho_i'$  are of dimension 1 and coincide along  $\gamma$ ; the existence of such a regular neighborhood (whose boundary is of dimension 2, so a surface) of  $\gamma$  gives us enough space to separate them so that they are disjoint. There exists an isotopic deformation around the path  $\gamma$  that maps the curves  $\rho_i'$  onto the boundary of this regular neighborhood. We thus obtain a new system of curves  $\{\rho_i \mid 1 \leq i \leq 2g\}$  that is isotopic to the first system  $\{\rho_i' \mid 1 \leq i \leq 2g\}$  via a continuous deformation. We thus obtain loops  $\rho_i$  that bound disks immersed in  $C^*$ , so it follows from Dehn's lemma 5.3.3 that the loops  $\rho_i$  bound disks  $D_i$  properly immersed in  $C^*$ . The Figures 5.5 and 5.6 perfectly illustrate this situation

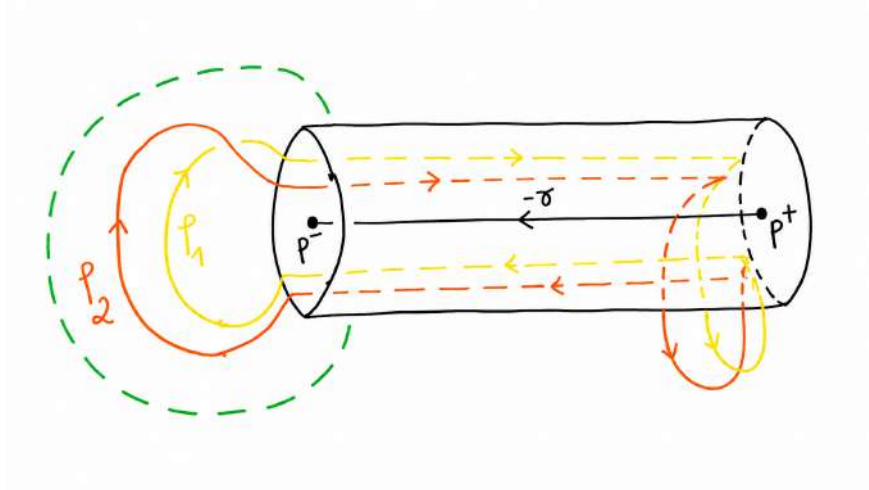


Figure 5.5: Curve separation

The area shaded in green is shown in the figure below 5.6 above,

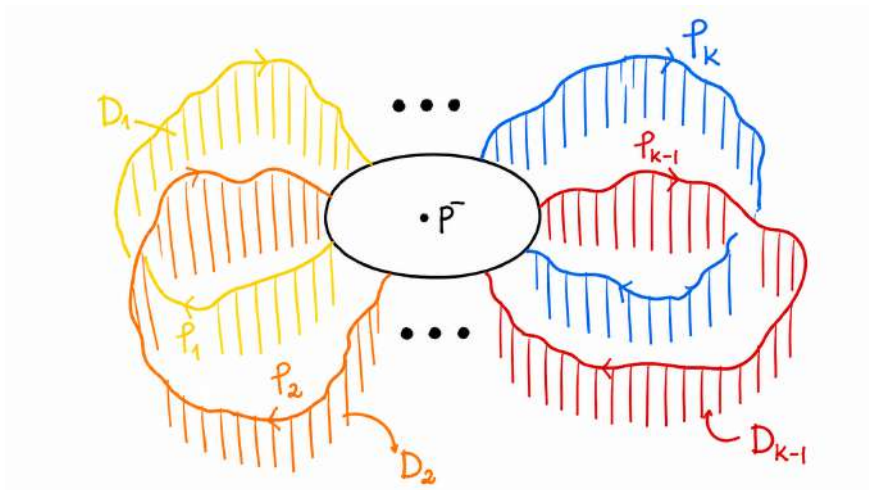
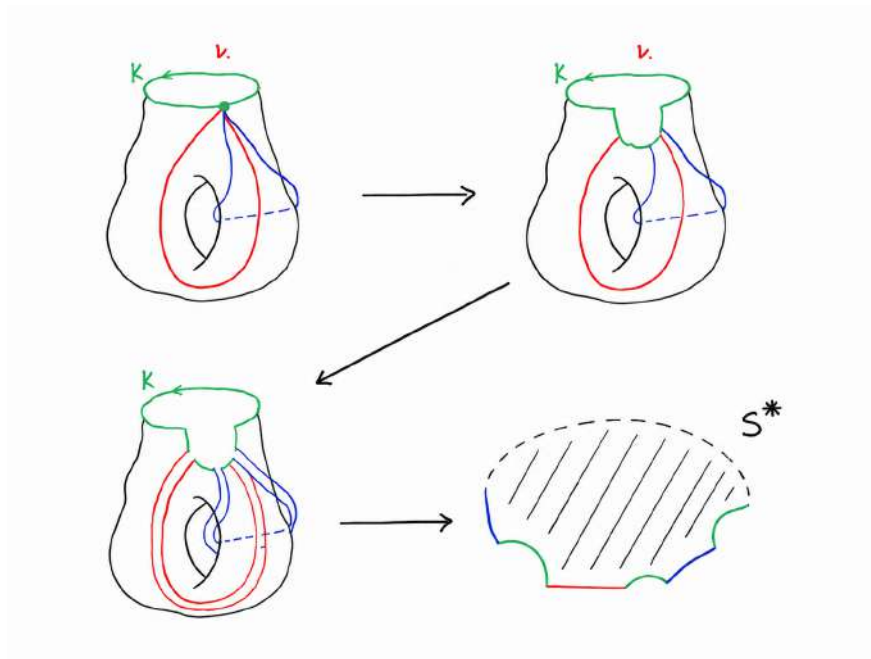


Figure 5.6: Preparation for cutting

Now, by cutting  $C^*$  along these disks  $D_i$ , we obtain a solid  $C^{**}$  of dimension 3 whose boundary is

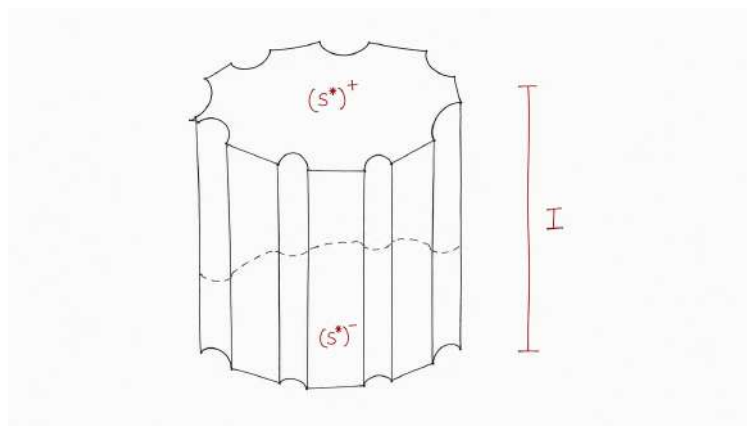
$$\partial C^{**} = (S^*)^+ \cup A \cup (S^*)^-$$

where  $S^*$  represents a disk,  $A = \partial S^* \times (0, 1)$  denotes a cylinder,  $(S^*)^+ = S^* \times \{1\}$  and  $(S^*)^- = S^* \times \{0\}$ .


 Figure 5.7: Construction of the disk  $S^*$ 

Topologically, the boundary  $\partial C^{**}$  resembles a surface of genus 0 with an empty boundary component; from the perspective of surface classification theory,  $\partial C^{**}$  is homeomorphic (in the abstract sense) to a sphere  $S^2$  embedded in  $S^3$ : thus, by Alexander's theorem 5.3.1,  $C^{**}$  is a 3-ball.

Note also from the above that we can visualize the boundary  $C^{**}$  as a cylinder, which naturally implies the existence of a fibered structure. More precisely, the topology of  $C^{**}$  induces a trivial fiber structure on the segment  $I$ , i.e.,  $C^{**} \cong S^* \times I$ . The figure 5.8 illustrates the situation,


 Figure 5.8: full solid  $C^{**}$ 

From the above, we have moved from the space  $C^*$  to construct a new space  $C^{**}$ , which is

equipped with a fiber bundle structure. It is natural to ask whether we can take the reverse path starting from the fiber bundle structure on  $C^{**}$ ?

From what we have just proved,  $C^{**}$  is a 3-ball, hence a 3-dimensional topological manifold whose boundary consists of  $(S^*)^- = S^* \times \{0\}$ ,  $(S^*)^+ = S^* \times \{1\}$ , and the cutting disks  $D_{i_1}$  and  $D_{i_2}$ .

We also know that  $C^* = C^{**}/x \sim g'_i(x)$  with  $x \in D_{i_1}$  and  $g'_i : D_{i_1} \longrightarrow D_{i_2}$  the (canonical) gluing homeomorphisms: such homeomorphisms always exist! On the other hand, the homeomorphisms  $g'_i$  do not necessarily preserve the trivial fibration on  $C^{**}$ .

Note that the trivial fibration structure on  $C^{**}$  induces a fibration structure on each of the cutting disks  $D_{ij}$ . Indeed: let  $\text{proj}_2 : C^{**} \cong S^* \times I \longrightarrow I$  denote the fibration<sup>2</sup> on  $C^{**}$ . The fibers of  $\text{proj}_2$  meet  $D_{ij}$  transversally in  $\gamma_{ij}$ , that is,  $D_{ij} \cap S^* = \gamma_j \cong I$  (\*).

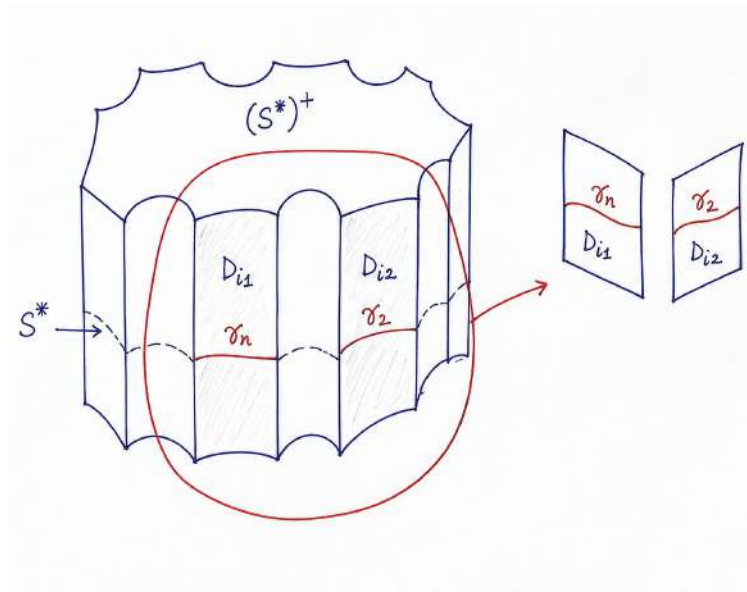


Figure 5.9: Fibration on  $D_{ij}$

Let us determine the fibration on  $D_{ij}$ .

Let  $\varepsilon > 0$  be such that  $I_\varepsilon = ]-\varepsilon; \varepsilon[$  is an open subset of  $I$ . It is clear that  $\text{proj}_2^{-1}(I_\varepsilon) \cong I_\varepsilon \times S^*$ . This in turn implies that

$$D_{ij} \cap \text{proj}_2^{-1}(I_\varepsilon) \cong I_\varepsilon \times C \quad \text{with } C = D_{ij} \cap S^* \cong I \quad (\text{by } (*)).$$

Hence

$$\text{proj}_2^{-1}|_{D_{ij}}(I_\varepsilon) \cong I_\varepsilon \times \gamma_{ij} \cong I \times I$$

Thus  $\text{proj}_2|_{D_{ij}} : D_{ij} = I \times I \longrightarrow I$  is a trivial fibration over  $I$  whose fiber is homeomorphic to  $I$ . By (\*), we have  $D_{ij} \cap (S^*)^\pm = I$ , and

<sup>2</sup>This is the second projection onto the unit interval  $I$ .

$$\begin{aligned}
& g'_i : D_{i1} \longrightarrow D_{i2} \text{ is a (gluing) homeomorphism} \\
\implies g'_i|_{D_{i1} \cap (S^*)^\pm} : D_{i1} \cap (S^*)^\pm & \longrightarrow g'_i(D_{i1} \cap (S^*)^\pm) \\
\implies g'_i|_{D_{i1} \cap (S^*)^\pm} : D_{i1} \cap (S^*)^\pm & \longrightarrow D_{i2} \cap (S^*)^\pm \\
\implies g'_i|_{D_{i1} \cap (S^*)^\pm} : I & \longrightarrow I \text{ is a homeomorphism fixing the endpoints 0 and 1} \quad (**).
\end{aligned}$$

Hence  $g'_i|_{D_{i1} \cap (S^*)^\pm} \in \text{Homeo}^+(I, \partial I) = \text{Homeo}^+(I, \{0, 1\})$ , and since the mapping class group of the unit interval is trivial, i.e.

$$\text{MCG}(I, \partial I) = \pi_0(\text{Homeo}^+(I, \{0, 1\})) = \{\text{Id}\}$$

it follows that all the  $g'_i|_{D_{i1} \cap (S^*)^\pm}$  are isotopic to the identity. More precisely, the homeomorphisms  $g'_i|_{D_{i1} \cap (S^*)^-}$  and  $g'_i|_{D_{i1} \cap (S^*)^+}$  are isotopic to the identity, and since the  $D_{ij}$  are products  $I \times I$ , we can extend these homeomorphisms throughout the interior of the disk  $D_{ij}$ .

Now let  $\varphi_t : I \longrightarrow I$ , with  $\varphi_0 = g'_i|_{D_{i1} \cap (S^*)^-}$  and  $\varphi_1 = g'_i|_{D_{i1} \cap (S^*)^+}$ , denote the isotopy between  $\varphi_0$  and  $\varphi_1$ . We set

$$g_i : D_{i1} \cong I \times I \longrightarrow D_{i2} \cong I \times I, \quad g_i(x, t) = (\varphi_t(x), t).$$

It is easy to see that

$$g_i(D_{i1} \cap (S^*)^-, 0) = (\varphi_0, 0) = (g'_i|_{D_{i1} \cap (S^*)^-}, 0)$$

We show in the same way that  $g_i(D_{i1} \cap (S^*)^+, 0) = (g'_i|_{D_{i1} \cap (S^*)^+}, 0)$  for any  $i$ . More precisely, this amounts to saying that

$$g_i|_{D_{i1} \cap (S^*)^-} = g'_i|_{D_{i1} \cap (S^*)^-}$$

It is clear that the  $g_i$  are homeomorphisms fixing the endpoints, by (\*\*). We have thus constructed homeomorphisms  $g_i : D_{i1} \rightarrow D_{i2}$  such that  $g_i|_{D_{i1} \cap (S^*)^-} = g'_i|_{D_{i1} \cap (S^*)^-}$  and which preserve the fibers (since  $g_i(I \times \{s\}) = I \times \{s\} \cong \text{proj}_2^{-1}(s)|_{D_{ij}}$ ).

Since

$$\text{MCG}(D_{ij}^2, \partial D_{ij}^2) = \pi_0(\text{Homeo}^+(D_{ij}^2, \partial D_{ij}^2)) = \{1\},$$



every orientation-preserving homeomorphism of  $D_{ij}$  fixing the boundary is isotopic to the identity. Therefore, for each  $i$ , the homeomorphisms  $g_i$  and  $g'_i$  are isotopic relative to

$$D_{i1} \cap (S^*)^+ \cup D_{i1} \cap (S^*)^-.$$

Hence the original identifications  $g'_i$  may be replaced, up to isotopy, by the fibre-preserving homeomorphisms  $g_i$ . Consequently, the quotient obtained from  $C^{**} \cong S^* \times I$  by gluing each pair  $D_{i1}$  and  $D_{i2}$  via  $g'_i$  is homeomorphic to the quotient obtained using the fibre-preserving maps  $g_i$ .

It follows that the product fibration on  $S^* \times I$  descends to the quotient space  $C^*$ . Therefore  $C^*$  is fibre-preserving homeomorphic to a mapping torus

$$S \times I/h,$$

for some homeomorphism  $h : S \rightarrow S$ . In particular,  $C$  fibres over  $S^1$  with fibre  $S$ .

□

## 5.3 Geometric interpretation of Stallings' theorem

In light of the preceding discussion, centered on Stallings' fibration theorem for the exteriors of knots, it is natural to seek to understand the geometric intuition behind this theorem in a sense, to grasp its geometric content. Beyond the algebraic criterion concerning the isomorphism induced by the inclusions of the push-out copies of a Seifert surface, this result has a particularly intuitive interpretation: the curves obtained by push-out in one direction can be connected to the curves obtained by push-out in the opposite direction via cylinders embedded in the complement of the knot.

The figure below, which depicts the pretzel knot  $p(1, -1, 3)$ , perfectly illustrates this idea.

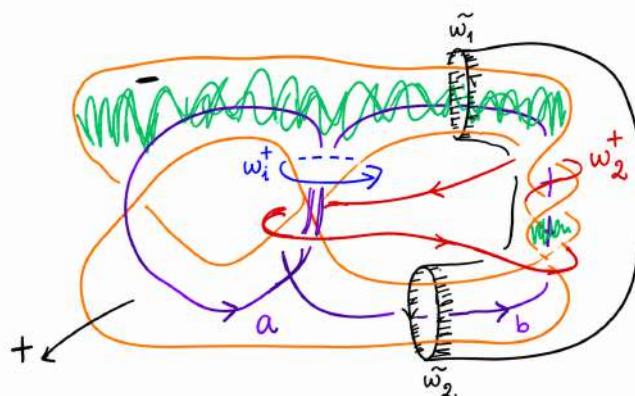


Figure 5.10: Pretzel knot  $p(1, -1, 3)$

I am aware that this figure 5.10 does not match the theory we have developed, since the knot involved is the trivial knot and the surface is of genus 1. And we know that the minimal genus surface for the trivial knot is of genus 0.

Now, since the pretzel knot  $p(1, -1, 3)$  is of genus 1, a basis of  $\pi_1(S)$  has two generating curves  $a = \omega_1$  and  $b = \omega_2$ .

By pushing these curves slightly in the positive direction of the normal to  $S$ , we obtain two new curves  $\omega_i^+$ , i.e.,  $\eta^+(\omega_i) = \omega_i^+$ . Similarly, a push in the negative direction yields two new curves  $\omega_i^-$ , i.e.,  $\eta^-(\omega_i) = \omega_i^-$ .

In the figure, the various loops  $w_1^+, w_2^+, w_1^-$ , and  $w_2^-$  represent such shifts of curves, all belonging to a basis of  $H_1(S)$ .

**Note:** Figure 5.10 is viewed from the perspective of homology, which is possible because, as we will see below, a homotopy isomorphism induces a homology isomorphism.

Recall that Stallings' fibration theorem for knot exteriors states that: a knot  $\mathfrak{k}$  is fibred over the circle if and only if the commutator subgroup  $\mathfrak{G}$  is of finite type. Equivalently, a knot  $\mathfrak{k}$  is fibred if and only if one of the two inclusions  $i_\pm : S^\pm \longrightarrow C^*$  induces a group isomorphism (at the homotopy level)

$$i_{\#}^\pm : \pi_1(S^\pm) \longrightarrow \pi_1(C^*)$$

By Lemma 2.1.11, we also obtain an isomorphism at the homology level

$$i_*^\pm : H_1(S) \longrightarrow H_1(C^*).$$

This property has a geometric consequence.

When the preceding isomorphism at the homology level holds, any homology class whose representative is a negative push can be expressed in a basis of  $H_1(C^*)$  as a linear combination of classes represented by positive pushes. More rigorously, suppose that the classes  $[\omega_1], [\omega_2], \dots, [\omega_g]$  form a basis for  $H_1(S)$ . Then each class  $[\omega_i^-]$  can be expressed in terms of the classes  $[\omega_1^+], [\omega_2^+], \dots, [\omega_g^+]$ , i.e.,

$$[\omega_i^-] = \prod_{j=1}^g \alpha_j [\omega_j^+]$$

**The geometric interpretation of this homological relation is given by the existence of a cylinder embedded in the complement of the knot, as visually represented in the figure.**

**Interpretation of Figure 5.10:** the curves  $\omega_1^-$  and  $\omega_2^-$  indicate the existence of a cylinder connecting these two curves to the positive normal  $\omega_2^+$ , and  $\omega_1^-$  and  $\omega_2^-$  co-bound this cylinder, i.e.,  $\partial(\text{cylinder}) = \omega_1^- \cup \omega_2^-$ .

**Conclusion:** The algebraic information of Stallings' fibration theorem, conveyed by the isomorphism (at the homology level), translates geometrically into the possibility of connecting the different pushes via families of cylinders located in the complement of the knot.

# Conclusion

Although Stallings's fibering theorem was proven by its author, John Stallings, in his original paper "*On fibering certain 3-manifolds*", the proof appears inaccessible at first glance, as it requires a very high level of abstraction. In this sense, this thesis was part of an extension to the third dimension within the framework of Stallings's fibration theorem, for a particular class of 3-manifolds: the exteriors of knots. The main objective of this thesis was to establish all the necessary algebraic, topological, and geometric tools to reproduce the proof of Stallings's fibration theorem for knot complements.

Throughout our study, we were guided by the main ideas developed in the book *Knots* by Burde and Zieschang [21]. First, we reviewed some classical results in algebraic topology, covering theory, and the geometry of 3-manifolds; we briefly presented knot theory as well as certain algebraic and geometric properties of knot complements. We then studied Seifert surfaces and the infinite cyclic covering of the knot complement, which are fundamental tools for understanding our work. Finally, we presented Stallings's fibration theorem for knot exteriors, emphasizing the relationships between the properties of the fundamental group and the existence of a fibration on the circle.

It follows that the study of fibred knots highlights a deep interaction between the topology of knot complements and the algebraic structure of their fundamental groups. More precisely, Stallings' theorem for knot complements provides a criterion for fibration on the circle based on finiteness properties of the derived subgroup of the fundamental group.

Although our analysis was conducted within a deliberately restricted framework that of knot complements this work is intended as a more intuitive approach to the more general theories concerning fibred 3-manifolds. Several much more advanced aspects have not been addressed here, notably issues related to the monodromy of fibrations, or to more sophisticated invariants such as the Alexander polynomial and its relations with fibred structures.

As this work is far from complete, several research avenues in modern topology can be explored. It would be appropriate, for the continuation of our work, to develop techniques allowing for a more in-depth study of the links between fibrations and knot invariants, to analyse certain classical examples of fibred knots such as the family of pretzel knots with arbitrary parameters, the study of which is underway in the recent work of Ana Garcia Lecuona, notably in her article "*Fibred ribbon pretzels*" [14], or to examine generalizations of Stallings' theorem in the more general setting of 3-manifolds.

# References

- [1] Ian Agol, *The virtual haken conjecture*, Documenta Mathematica **18** (2013), 1045–1087.
- [2] James W. Alexander, *On the subdivision of 3-space by a polyhedron*, Proceedings of the National Academy of Sciences **10** (1924), no. 1, 6–8.
- [3] Cheryl Balm, Stefan Friedl, Efstratia Kalfagianni, and Mark Powell, *Cosmetic crossings and seifert matrices*, arXiv preprint arXiv:1108.3102 (2011).
- [4] Martin R. Bridson, Dawid Kielak, and Monika Kudłinska, *Stallings’s fibering theorem and  $PD^3$ -pairs*, arXiv preprint arXiv:2307.10725 (2023).
- [5] G. Burdes and H. Zieschang, *Knots*, Walter de Gruyter, Berlin and New York, 2003.
- [6] Benson Farb and Dan Margalit, *A primer on mapping class groups*, Princeton Mathematical Series, vol. 49, Princeton University Press, 2012.
- [7] Yves Félix and Daniel Tanré, *Topologie algébrique: cours et exercices corrigés*, Dunod, 2010.
- [8] Satoshi Fukuhama, Makoto Ozawa, and Masakazu Teragaito, *Genus one, three-bridge knots are pretzel*, Journal of Knot Theory and Its Ramifications **8** (1999), 879–886.
- [9] Claude Godbillon, *éléments de topologie algébrique*, Hermann, 1971.
- [10] André Gramain, *Topology of surfaces*, BCS Associates, 1984.
- [11] Allen Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge, 2002.
- [12] John Hempel, *3-manifolds*, Princeton University Press, 1976.
- [13] Eriko Hironaka, *The lehmer polynomial and pretzel links*, Canadian Mathematical Bulletin **44** (2001), no. 4, 440–451.
- [14] Ana G Lecuona and Andy Wand, *Fibered ribbon pretzels*, Bulletin of the London Mathematical Society **58** (2026), no. 1, e70271.
- [15] Charles Livingston, *Knot theory*, vol. 24, Cambridge University Press, 1993.
- [16] Jakob Nielsen, *Untersuchungen zur topologie der geschlossenen zweiseitigen flächen*, Acta Mathematica **50** (1927), 189–358.
- [17] Dale Rolfsen, *Knots and links*, American Mathematical Society, Providence, RI, 2003.
- [18] Martin Scharlemann, *Topology of knots*, Topological aspects of the dynamics of fluids and plasmas, Springer, 1992, pp. 65–82.
- [19] Nancy Scherich, *The alexander polynomial*, (2013).
- [20] John Stallings, *On the loop theorem*, Annals of Mathematics **72** (1960), no. 1, 12–19.
- [21] ———, *On fibering certain 3-manifolds*, Topology of 3-manifolds and related topics (1962), 95–100.

# Appendix

## Some Fundamental Results in the Geometric Topology in dimension 3

In this section, we review some classical results in the third dimension that are used in Chapter 5.

**Theorem 5.3.1** (Alexander's Theorem). *Every 2-sphere embedded in  $S^3$  is a boundary of a 3-sphere.*

This is a combinatorial version of Schoenflies' sphere theorem in 3 dimensions.

*Proof.* We can find this in Alexander's original paper [2]. □

**Theorem 5.3.2** (Nielsen's Theorem). *Let  $S$  be a closed, orientable surface of genus  $g \geq 1$ , and let  $\varphi : H_1(S; \mathbb{Z}) \rightarrow H_1(S; \mathbb{Z})$  be an isomorphism of abelian groups. We assume that  $\varphi$  preserves the intersection form, that is,*

$$\langle \varphi(\alpha), \varphi(\beta) \rangle = \langle \alpha, \beta \rangle \quad \forall \alpha, \beta \in H_1(S; \mathbb{Z}),$$

*in other words,  $\varphi \in \text{Sp}(2g, \mathbb{Z})$ . Then there exists a homeomorphism  $f : S \rightarrow S$  such that*

$$H(f) = \varphi.$$

*Furthermore, such a homeomorphism is unique up to isotropy.*

*Proof.* See [16]. □

**Theorem 5.3.3** (Dehn's Lemma). *Suppose  $M$  is a 3-manifold and  $f : B^2 \rightarrow M$  is a map such that for some neighborhood  $A$  of  $\partial B^2$  in  $B^2$ ,  $f|_A$  is an embedding and  $f^{-1}(f(A)) = A$ . Then  $f|_{\partial B^2}$  extends to an embedding  $g : B^2 \rightarrow M$ .*

Simply put, this theorem tells us that if a loop defined along the boundary of a topological 3-manifold  $M$  bounds a disk immersed inside  $M$ , then it also bounds a properly embedded disk in  $M$ .

*Proof.* For a complete and detailed proof, the reader may consult John Hempel's famous book "3-manifolds," pages 39–54 [12] □

**Theorem 5.3.4** (Loop Theorem [20], Section 2). *Let  $M$  be a 3-manifold and let  $B$  be a component of its boundary. Let  $N$  be a normal subgroup of  $\pi_1(B)$ ; suppose there are elements in  $\ker[\pi_1(B) \rightarrow \pi_1(M)]$  that are not in  $N$ . Then there is a simple loop  $C \subset B$  such that  $C$  bounds a nonsingular disk in  $M$ , and such that  $C(B) \notin N$ .*

*Proof.* For technical details, see John Stallings' article [20]. □